



Sign & Spell VR – Irish Sign Language Alphabet Practice Final Project Report

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Abstract

The Sign & Spell VR project aims to create an immersive virtual reality application designed to facilitate the learning of the Irish Sign Language (ISL). This project addresses the need for accessible and interactive tool for sign language education, focusing specially on ISL, which is less commonly represented in existing digital learning platforms. The application utilizes the Meta Quest 3/3S virtual reality headset and the Godot 4 game engine to provide an engaging and interactive learning environment.

The core functionality of the application includes real-time tracking of user's hand movements to recognize and assess Irish Sign Language alphabet letters signed by the user. This features immediate feedback to aid learning effectiveness and user engagement. The project progresses through two main stages: a learning module that teaches users how to sign each letter of the Irish Sign Language alphabet, followed by a testing phase that evaluates how well they can perform the signs on their own.

The final deliverable is a functional VR prototype that demonstrates the potential of virtual reality technology in language learning and accessibility. By integrating gesture recognition with VR immersion, the project offers a novel approach to learning ISL, which benefits both learners and the deaf community by promoting effective communication skills. The initiative not only supports language acquisition but also raises awareness about the cultural significance of ISL. The Sign & Spell VR project thus stands as an innovative educational tool with possibilities for further expansion to include additional sign language and more complex linguistic features.

Declaration

I hereby declare that the work described in this dissertation is, except where otherwise stated, entirely my own work and has not been submitted as an exercise for a degree at this or any other university.

Signed:

*Karl Negrillo*_____

Karl Negrillo

Date

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1. Introduction

Irish Sign Language (ISL) plays an important role in communication and accessibility within Ireland, particularly for members of the Deaf community. However, learning ISL as a beginner can be quite difficult, especially without access to regular feedback or structured guidance. Many learners rely on resources such as books or videos, which can be helpful, but these often lack interactivity. Because of this, it can be hard to know whether a sign is being performed correctly.

With the increasing use of interactive technologies, virtual reality (VR) has started to be explored as a possible learning tool. VR allows users to engage with content in a more active way, rather than just observing it. In theory, this could make learning ISL more practical, although it also introduces new challenges in terms of usability and accuracy.

This project looks at how VR can be used to support ISL learning, with a focus on creating something that feels approachable for beginners rather than overly complex.

1.1. Project Background

Learning Irish Sign Language can be challenging, particularly for beginners who are practising on their own. While there are many resources available, such as videos or online tutorials, these do not always provide immediate feedback. This means that learners might repeat incorrect signs without realising it, which can slow down progress over time.

Virtual reality offers a different way of approaching this problem. Instead of watching demonstrations, users can actively practise signs within a simulated environment. This seems especially useful for sign language, where learning depends heavily on hand shapes and movement rather than just memorising information.

Some existing VR applications have already explored sign language learning, but most of them focus on American Sign Language. This makes it harder for learners who are specifically interested in ISL, as there are fewer dedicated tools available. Because of this, there appears to be a gap in more localised learning systems.

This project was developed with that in mind. The idea was to create a simple VR application that allows users to practise ISL letters while receiving immediate feedback. While this does not fully solve all learning challenges, it aims to make the process more interactive and slightly more engaging than traditional methods.

1.2. Project Description

Sign & Spell VR is a virtual reality application designed to help users learn the Irish Sign Language (ISL) alphabet in a more interactive way. The system runs on the Meta Quest 3/3S headset and is built using the Godot 4 game engine. It uses hand tracking so that users can perform signs naturally, rather than relying on controllers.

The application is split into two main parts. First, there is a learning section where users are introduced to individual ISL letters and can practise them. After this, there is a testing section where users attempt to recall and perform the signs they have learned.

When a user performs a sign, the system compares their hand movement with predefined ISL gestures and provides feedback. In most cases, this is a simple response such as confirming whether the sign is correct or asking the user to try again. While this feedback is useful, it is also quite basic, mainly due to the limitations of hand tracking accuracy.

Originally, the project aimed to include more advanced features, such as learning full words or simple sentences. However, during development it became clear that this would be difficult to achieve reliably within the available time. Because of this, the scope was reduced to focus on the ISL alphabet. Although this makes the system more limited, it also means that the core functionality could be developed in a more stable way.

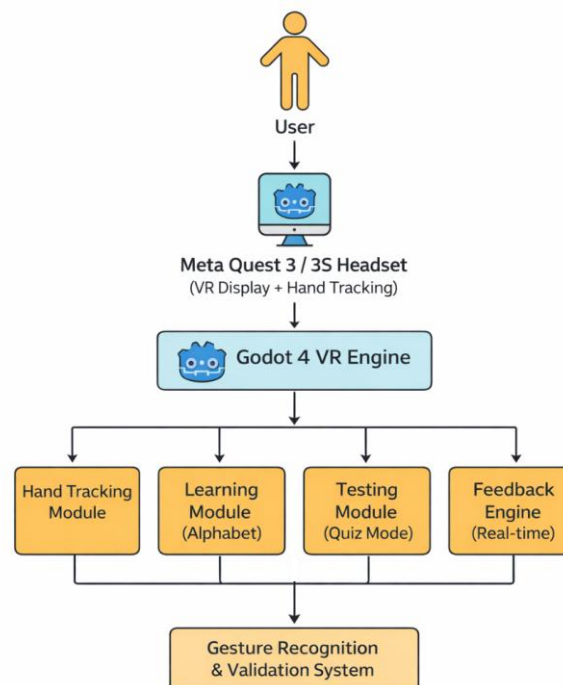


Figure 1: System Architecture Diagram for Sign & Spell VR

1.3. Project Aims and Objectives

The main aim of this project is to design and develop a virtual reality application that supports the learning of Irish Sign Language through interactive practice.

To achieve this, several objectives were identified:

- To integrate hand tracking using the Meta Quest 3 headset

- To represent the ISL alphabet within a virtual environment
- To create a space where users can practise individual signs
- To develop a simple testing section to assess learning
- To allow users to repeat and revisit content as needed
- To gather feedback on usability and overall experience

These objectives are relatively straightforward, but they reflect the focus of the project, which is to build something functional and usable rather than overly complex.

1.4. Project Scope

The project scope focuses on the VR application. It clarifies the specific features and deliverables that will be developed, as well as what falls outside the project's responsibilities. By setting clear limits, the project scope ensures that work remains targeted, achievable within available resources and time, and helps manage expectations for stakeholders and users.

In Scope:

- **ISL Learning Modules** – Users learn about the ISL alphabet, practice the ISL alphabet, and can test themselves in an interactive VR Environment.
- **Real-Hand Tracking & Feedback** – The system tracks hand movements and provides immediate visual feedback to guide users and correct errors.
- **Progress Assessment** – Users can test their knowledge and review feedback to evaluate learning and identify areas of improvement
- **Intuitive VR Interface** – The application is designed for the Meta Quest 3/3S headset, ensuring a comfortable and accessible user experience
- **Single-User Mode** – The learning experience is focused on individual progress, with users able to repeat modules and revisit lessons as needed.

Out of Scope:

- Support for other VR platforms
- Learning beyond the ISL alphabet, such as words or sentences
- Multi-user or collaborative features
- Advanced tracking improvements or AI-based recognition

While some of these features were considered, they were not included due to time and technical constraints. This reflects a decision to focus on building a smaller, working prototype instead of a more ambitious system that might not be completed.

1.5. Thesis Roadmap

This report is structured to provide a comprehensive overview of my project Sign & Spell VR. The following chapters are included:

- **Literature Review** – This chapter will explore existing solutions, relevant technologies, and related research, providing an evaluation that informs the project direction.
- **System Analysis** – This chapter will outline the system requirements, including gathering and analysing user needs, and specifying system objectives
- **System Design** – In this chapter, it will be about the selected software development methodology, system architecture, and user interface design.
- **Testing and Evaluation** – This chapter will cover the testing strategies used, user evaluation, and results that assess the system's usability and effectiveness.
- **Issues and Future Work** – Finally this chapter reflects on the challenges faced during the project and proposed direction for future improvements.

2. Literature Review

2.1. Introduction

This chapter presents the research conducted to inform the development of the Sign & Spell VR project. It provides an overview of existing solutions in the realm of sign language education, as well as technologies relevant to learning tools that leverage immersive technologies like Virtual Reality (VR). The chapter will analyse these solutions to evaluate their strengths, weaknesses, and applicability to this project.

A core focus will be on exploring and comparing various technologies for key components such as hand tracking, user interaction, and immersive feedback - all essential for creating an effective learning environment in VR. The research will also delve into broader educational theories and methodologies, with a focus on accessible learning and the community's needs.

Additionally, this section will review similar final year projects and studies related to VR-based learning systems, highlighting how their findings have directly influenced the design choices and overall direction of Sign & Spell VR. By synthesizing these insights, this chapter aims to establish a theoretical foundation for the project and explain how prior work has shaped its development.

2.2. Alternative Existing Solutions

2.2.1 Introduction

There are now a wide range of applications available for learning sign language, each taking a slightly different approach. Some focus on structured lessons, while others use games or interactive environments to make learning more engaging. Because of this variation, it can be difficult to compare them directly without a clear structure.

To address this, the FURPS model was used as a simple evaluation framework. This includes Functionality, Usability, Reliability, Performance, and Supportability. While it does not cover everything in detail, it provides a useful way of identifying what each system does well and where it may fall short.

For this project, a number of existing solutions were reviewed, including ASL Fingerspeller, Silent Classroom VR, ASL Champ!, DEF-ISL, and a VR hand tracking demonstration. These were selected because they represent a mix of VR and non-VR approaches, as well as different levels of complexity. Each system is considered in terms of how it supports learning, its strengths and weaknesses, and how it influenced the design of Sign & Spell VR.

2.2.2 ASL Fingerspeller – SideQuest

ASL Fingerspeller is a VR app available on SideQuest that lets users practice fingerspelling in American Sign Language (ASL) using the Meta Quest headset. The app allows learners to practice spelling names, places, and random words, with a timed mode that adds some challenge to the experience. The app's store description claims that it helps users learn the ASL alphabet quickly and easily.

From the Functionality aspect, ASL Fingerspeller does a good job of teaching the alphabet. However, user feedback reveals that the app can become repetitive over time, particularly when users are only practicing the alphabet. While repetition is effective for reinforcing the ASL alphabet, it doesn't offer much Variety or Progression, which could make it harder for learners to stay engaged after mastering the basics.

For Sign & Spell VR, the main takeaway from ASL Fingerspeller is the idea of incorporating challenge elements, such as quiz modes or more dynamic lessons. A gradual expansion from simple alphabet practice to everyday words and even basic sentences could help maintain the learner's engagement and motivation. This would ensure that the learning experience doesn't become monotonous and provides learners with more opportunities to apply their knowledge in real-world contexts [1].

2.2.3 Silent Classroom VR – Meta

Silent Classroom VR is a classroom-based VR experience designed to teach American Sign Language through a simulated classroom environment on the Meta Quest. It immerses learners in a virtual classroom where they can practice hand signs via hand tracking. The app's description highlights the use of hand tracking to reinforce finger-spelling skills, and user reviews often praise the app's immersive nature, stating that it feels like being in a real ASL classroom.

While Silent Classroom VR excels in creating an immersive experience, it primarily focuses on ASL, not ISL, and its reliance on a classroom setting may not be ideal for users who prefer shorter, focused practice sessions. The app is designed for learners who enjoy structure and prefer a more formal learning environment. However, for those just beginning to learn sign language or those looking for more casual practice, a full classroom environment might feel overwhelming or unnecessary.

For Sign & Spell VR, this suggests that while immersion is important for making learning feel more authentic, the system should prioritize quick practice loops and immediate feedback to better serve ISL beginners. A simplified and focused environment would be more suitable for beginners who may find traditional classroom settings too rigid. This ensures that the experience is accessible, without overwhelming users who are new to both sign language and VR technology [2].

2.2.4 ASL Champ! – Science Direct

ASL Champ! is another VR-based learning tool that uses AI-based sign recognition to provide real-time feedback on users' signs. According to the ScienceDirect abstract, the app gives users immediate feedback, helping them understand whether their sign is correct or not. This real-time feedback is a key feature of ASL Champ!, making it easy for users to track their progress and correct mistakes on the spot.

This tool demonstrates the importance of immediate feedback in supporting the learning process, especially when it comes to sign language, where correct hand shapes and positions are crucial. While Sign & Spell VR does not currently utilize AI for sign recognition, this research inspired the decision to incorporate a rule-based real-time feedback system. Even without AI, Sign & Spell VR aims to provide users with clear, instant feedback (correct/incorrect, suggestions) to keep them engaged and motivated.

For Sign & Spell VR, this finding reinforces the importance of feedback loops in motivating users, even without the complexities of AI. It also highlights the need for accuracy and clarity when delivering feedback, ensuring users have the support they need as they progress in their learning [3].

2.2.5 DEF-ISL – App Store

DEF-ISL is an app for learning Indian Sign Language (ISL). While this app is not focused on Irish Sign Language (ISL), it offers a good example of how sign language apps can incorporate video demonstrations and interactive lessons. DEF-ISL is designed to be easy to use and suitable for all ages, featuring a variety of signs and video resources to help learners get started.

What's notable about DEF-ISL is its emphasis on learning through visual resources. However, it still lacks active practice elements, such as hand tracking or real-time feedback, which can make the learning process feel passive. Sign & Spell VR seeks to address this gap by offering interactive practice with real-time feedback. This allows learners to receive corrections immediately, improving their retention and confidence. Although DEF-ISL is helpful for learning Indian Sign Language, Sign & Spell VR offers a more dynamic learning environment focused on hands-on practice rather than passive observation [4].

2.2.6 Design of Sign Language Learning Media Based on Virtual Reality

In a study by Int. J. Online Biomedical Engineering, the authors describe the design of a VR-based sign language learning application targeted at students with special needs. The app combines interactive 3D environments with visual and audio feedback, helping users feel more engaged in the learning process. According to the study, users reported increased confidence in using sign language after participating in the VR sessions, which underscores the importance of immersion and feedback in improving user engagement.

This finding directly supports the design decisions made in Sign & Spell VR, particularly the focus on creating an immersive environment where learners can practice signs and receive real-time feedback. By incorporating interactive practice and confidence-building activities, Sign & Spell VR aims to offer an engaging experience for ISL learners, inspired by the principles Int. J. Online Biomedical Engineering research. The goal is to provide learners with an environment where they can practice at their own pace while receiving clear, actionable feedback [5].

2.2.7 VR Sign Language Learning with Hand Tracking – YouTube

A YouTube video titled “VR Sign Language Learning with Hand Tracking” showcases how Meta Quest hand tracking can be used in VR without requiring controllers. The video emphasizes that hand tracking enables a more natural interaction, where learners can use their real hands to practice signs. This feature is pivotal for making the learning process feel more authentic and intuitive.

This aligns directly with the goals of Sign & Spell VR, which leverages controller-free interaction to create a more intuitive learning experience. By focusing on hand-based input, Sign & Spell VR aims to provide a seamless, user-friendly interface that mimics natural sign language interactions. This approach is especially beneficial for learners new to both sign language and VR technology, as it makes the system more accessible and easy to use [6].

2.2.8 Existing Solutions Conclusion

This section has provided a critical analysis of several existing solutions using the FURPS model. It highlights the strengths of each tool, such as immersion and real-time feedback, while also identifying gaps in ISL specific learning, real-time feedback accuracy, and engagement. Sign & Spell VR aims to improve on these existing solutions by offering a more focused, interactive experience tailored to ISL beginners. The findings from these existing solutions have informed the development of Sign & Spell VR, particularly in terms of feedback loops, hand tracking, and simplicity.

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2.3. Technologies Researched

For this project, several technologies were investigated to determine which would be most suitable for building a VR sign language learning application. Popular engines such as Unity and Unreal Engine were considered, but the Godot 4 Engine was ultimately selected because it is free, open source, and widely regarded as approachable for developers with limited game-development experience [7]. It is also increasingly popular among indie developers, and there is a growing set of tutorials specifically focused on VR development in Godot. In addition, Godot 4 aligns with the XR Prototyping module used in the course, which makes it a practical choice for both learning and implementation.

2.3.1 Meta Quest 3/3S

The Meta Quest 3 headset was chosen as the target platform. It functions as a standalone device and does not require a high-end gaming PC, which simplifies setup and makes it more accessible for classroom or home use. Crucially, the Quest 3 supports built-in hand tracking, allowing accurate capture of finger movements without controllers. This is important for a sign language learning app because it enables more precise and natural representation of ISL hand shapes. The wireless design allows users to move freely while practising, reducing complexity in the overall system.

A substantial number of guides and examples exist for developing educational and training applications on Meta Quest, which is helpful given the learning curve of VR development. The device is also increasingly used in educational settings, reinforcing its suitability as a platform for Sign & Spell VR. [14]



Figure 2: Meta Quest 3/3S

2.3.2 Godot 4 Engine

The project uses the Godot 4 game engine as its primary development environment. Godot is free and open source, removing licensing barriers and making it suitable for a student project. Although it is less widely used than Unity or Unreal, recent versions include significantly improved support for VR through OpenXR, which is essential for Meta Quest integration.

Godot 4 supports full 3D environments and provides tools that are well-suited to building educational and experimental projects. Because the project centres on creating an environment where users can practise ISL and receive feedback, Godot's scene system, scripting model, and VR plugins are sufficient to implement the required features. Its open-source nature also means there is an active community producing tutorials, examples, and answers to common problems. This is an important factor when learning new XR techniques. [8]

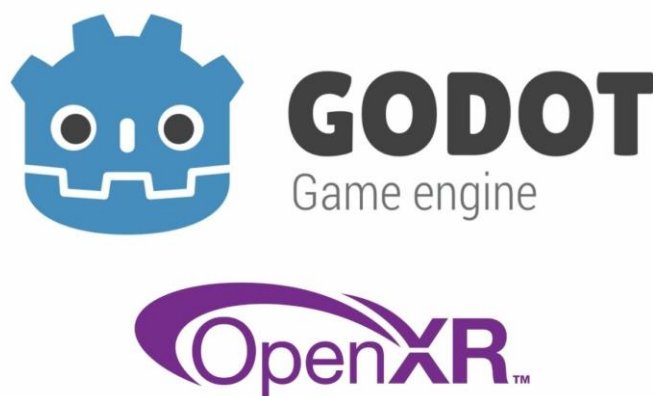


Figure 3: Godot and OpenXR

2.3.3 GDScript

Most of the project's logic is implemented in GDScript, Godot's built-in scripting language. GDScript has a syntax similar to Python, which makes it easier to learn and read, especially for developers with some prior Python experience. Alternative languages such as C# are supported, but they typically require more configuration and are less tightly integrated into the engine's workflow.

Using GDScript allows rapid iteration and testing when implementing features such as hand-movement checks, correctness feedback, and saving user progress. This reduces overhead when experimenting with new ideas or fixing bugs, which is particularly valuable in a project that involves frequent changes to interaction logic and UI behaviour. [8]

2.3.4 OpenXR

The project uses OpenXR with Godot 4 as the foundation for VR support on Meta Quest 3. OpenXR is an open standard that defines how VR hardware and applications communicate, allowing the application to work more smoothly across compatible devices. Godot 4 includes native OpenXR support, so activating the plugin and following available guides is sufficient to enable core VR features.

OpenXR provides access to essential capabilities such as head tracking and hand tracking without requiring custom low-level integrations. This allows the hand-tracking module in Sign & Spell VR to obtain real sensor data from the Quest and feed it directly into the sign-recognition logic during practice sessions. It also contributes to a smoother, less glitchy VR experience, which is important for beginners who may be sensitive to latency or instability. By following the official documentation and examples, VR rendering and tracking could be enabled relatively quickly, allowing more development time to focus on ISL-specific features rather than platform integration. [8]

2.3.5 Blender

The project uses Blender as a supporting tool for 3D asset creation. Blender is free and open source, removing licensing barriers and making it suitable for a student project. Although it has a steeper learning curve than simpler 3D tools, recent versions include significantly improved tools for precise modelling, which is essential for creating accurate ISL fingerspelling hand shapes.

Blender supports detailed mesh editing and provides tools that are well-suited to building educational assets like hand models for sign language practice. Because the project centres on creating accurate 3D representations of ISL alphabet letters, Blender's sculpting, rigging, and UV mapping features are sufficient to implement the required hand models. Its open-source nature also means there is an active community producing tutorials, examples, and answers to common modelling problems. This is an important factor when learning 3D techniques for VR projects. [19]



Figure 4: Blender

2.3.6 Justification of Technology Selection

2.3.6.1 Why not Unity or Unreal Engine?

Unity was considered as a potential engine because it is widely used for VR and offers many built-in features and assets. However, the overall setup was judged to be less beginner-friendly, especially for quickly enabling hand tracking and OpenXR-based VR. The configuration process can be complex, with many options and systems that are not directly relevant to this project's scope. Unity's licensing model can also introduce limits or costs when using certain features, which is less suitable for a student project.

Given that the associated XR Prototyping course focuses specifically on Godot for VR and hand tracking, it was more efficient and consistent to use the same engine in this project. This alignment avoids the need to learn a completely new toolchain and allows development time to be spent implementing and testing features rather than managing engine-specific setup.

Unreal Engine is highly capable, particularly for high-end games and graphically intensive experiences. However, most available guidance and examples are aimed at large projects or experienced developers. For a smaller educational prototype, Unreal would introduce unnecessary complexity, focusing more on advanced rendering and less on rapid iteration for hand-tracking interactions. As a result, it was considered excessive for the needs of Sign & Spell VR, where the emphasis is on functionality, feedback, and accessibility rather than cutting-edge graphics.

2.3.6.2 Why not a Database

Relational or cloud databases (such as SQL or Firebase) were also considered for storing user accounts and calibration information. After analysis, this approach was deemed more complex than necessary for the expected scale of the application. The system is not designed for thousands of concurrent users; instead, it primarily needs to store configuration and progress data for individual learners on their own devices.

JSON files integrate cleanly with Godot's FileAccess API, require minimal setup, and are easy to inspect and debug during development. This keeps the architecture simple while satisfying the requirement for persistent local storage. If the project later expands to support multi-user scenarios or centralised analytics, the JSON data model can still be migrated to a database-backed solution, but for the current prototype, file-based storage is the most practical choice.

2.4. Other Relevant Research

2.4.1 Ethics

Ethics plays a fundamental role in the development of sign language learning technologies, especially when it comes to representing and interacting with Deaf communities. Technologies like AI-driven sign language recognition or virtual reality (VR) applications require thoughtful consideration of cultural sensitivity, linguistic diversity, and informed consent. As these systems increasingly collect and utilize sign language data, often involving personal and culturally significant expressions, the importance of maintaining ethical standards in development becomes clear.

A key issue arises from the fact that sign languages are not just a means of communication but are also deeply intertwined with the cultural identities and practices of the Deaf community. Without

appropriate consultation, technological advancements risk misrepresenting or oversimplifying these languages, which could lead to harmful stereotypes or misuse of data. This concern is especially pronounced when AI models are used to recognize or generate signs, as they may fail to capture the nuances of sign language grammar and cultural context.

As we explore existing research and ethical concerns in this section, we will discuss the importance of community involvement in developing such systems, the ethical implications of using sign language data, and how these concerns directly influence the design choices made in Sign & Spell VR. It is essential that technologies built for language learning are not only effective but also respectful, ensuring that they serve the educational needs of users while also safeguarding cultural and linguistic integrity.

2.4.2 The FATE Landscape of Sign Language AI Datasets

One of the key studies in this area is TACCESS research on the FATE landscape (Fairness, Accountability, Transparency, and Ethics) of sign language AI datasets. The authors emphasize that signing videos used to train AI systems are not only technical data but also culturally rich and personal expressions of identity. As a result, these datasets must be handled with great care, ensuring informed consent from Deaf participants. The paper points out the risks of cultural appropriation, where the development of AI technologies is done without community consultation, leading to mistrust and misrepresentation.

For Sign & Spell VR, this paper underscores the need to consult with the Deaf community throughout development to ensure that the ISL signs included in the app are accurate, respectful, and culturally appropriate. As a result, feedback from Deaf users will be crucial in shaping both the content and interface of the app, ensuring that Sign & Spell VR does not only function as an educational tool but also as a culturally sensitive resource. Moreover, the project emphasizes ethical data usage and transparency in its design choices, especially in how hand tracking data is handled [9].

2.4.3 Sign Language Recognition and Interdisciplinary Challenges

The paper by Bragg et al. discusses the interdisciplinary challenges in developing AI systems for sign language recognition, particularly the ethical dilemmas that arise when the Deaf community is not consulted during the development process. The authors argue that sign languages are complex, involving not just hand gestures but also facial expressions and body movements, which can vary widely between communities. They highlight that technology often fails to respect linguistic identity, leading to misrepresentation and harmful stereotypes.

This research is highly relevant for Sign & Spell VR as it informs the decision to avoid relying on AI-based sign recognition for now, as it might misinterpret signs or fail to capture the nuances of ISL. Instead, the app uses a rule-based system for hand tracking and feedback, ensuring that signs are interpreted consistently and accurately. This decision aligns with ethical principles by focusing on accuracy and respect for the linguistic integrity of ISL [10].

2.4.4 AI Fairness for Deaf or Hard of Hearing (DHH) Users

SIGACCESS discuss the fairness of AI systems designed for Deaf or Hard of Hearing (DHH) users. The paper highlights that many AI systems perform poorly with Deaf users, especially due to limited or

biased data. The lack of transparency in AI models also makes it difficult for users to understand how their data is used or how decisions are made, leading to inequities in access and outcomes. The authors advocate for inclusive data collection and evaluation metrics that are specifically designed to cater to DHH needs.

For Sign & Spell VR, this finding aligns with the decision to use real-time feedback rather than AI-driven models. The feedback in Sign & Spell VR is designed to be transparent and easy to understand, helping users correct their signs in real-time and improve accuracy. The focus on real-time interaction instead of AI recognition minimizes the risk of biased feedback and ensures that learners are supported throughout the process [11].

2.4.5 WFD and WASLI Statement on Signing Avatars

The World Federation of the Deaf (WFD) and the World Association of Sign Language Interpreters (WASLI) have made it clear that signing avatars should not replace human interpreters. In their statement, they assert that avatars cannot replicate the full complexity of sign languages, especially the grammar, cultural nuances, and contextual meaning that are inherent in face-to-face communication. They recommend that avatars should only be used for non-urgent content, and always in consultation with the Deaf community.

For Sign & Spell VR, this research reinforces the decision not to use signing avatars for interpretation within the app. Instead, the focus is placed on real-time feedback based on accurate hand tracking. This decision ensures that human expertise remains at the core of the learning process, and the app respects the linguistic complexity and cultural significance of ISL. By avoiding avatars and focusing on real-time, learner-driven feedback, Sign & Spell VR stays true to the principle that human interaction is central to effective sign language learning [12].

2.4.6 Designing Strategic and Interactive Signing Instruction (SISI)

The SISI (Strategic and Interactive Signing Instruction) framework, as outlined in the paper "Designing Strategic and Interactive Signing Instruction", offers a comprehensive approach to teaching sign language, which directly informs the development of Sign & Spell VR. This framework incorporates proven teaching strategies to foster signed literacy skills through interactive learning cycles, and its focus on gradual skill-building, paired with visual aids and feedback, aligns well with the goals of the Sign & Spell VR project.[20]

Key Elements of SISI and Their Application to Sign & Spell VR:

1. Model:

- **SISI:** Instruction begins with the demonstration of the sign, followed by scaffolding where learners observe and replicate the gesture.
- **Sign & Spell VR:** In Sign & Spell VR, this model translates into the use of avatar models that demonstrate the correct hand shapes for each ISL sign. These avatars will provide clear visual guidance to learners, especially during the calibration phase where users can refer to the hand shape they are trying to mimic.

2. Deconstruct:

- **SISI:** Learners break down the complex gestures into simpler components to understand the nuances of the signs.
- **Sign & Spell VR:** This phase is supported in the VR environment by isolating parts of each sign and utilizing a skeleton overlay. As learners perform their signs, they will receive real-time feedback highlighting areas of the gesture where improvement is needed (e.g., mismatches in hand positioning).

3. Guided Practice:

- **SISI:** Learners practice the signs with prompts and receive immediate corrections from the instructor.
- **Sign & Spell VR:** This mirrors the system used in Sign & Spell VR, where learners practice signs and receive instant visual feedback (e.g., red or green signals) to correct errors immediately. This not only enhances accuracy but also keeps learners engaged and motivated through active learning.

4. Independent Practice:

- **SISI:** Once learners have practiced enough, they are encouraged to perform independently.
- **Sign & Spell VR:** The system's free practice mode aligns with this phase of the SISI framework. Users can independently practice the signs at their own pace, with progress tracked via the system. The JSON tracker helps monitor user improvements over time, providing them with a detailed view of their learning progress.

5. Share/Assess:

- **SISI:** Learners are encouraged to share their progress and engage in self-assessment or peer review to reinforce what they've learned.
- **Sign & Spell VR:** The final component of the SISI framework is incorporated into Sign & Spell VR by enabling learners to revisit previous lessons, review their performance, and receive revision prompts. Users can test their knowledge through assessments that help them measure their improvement and revise any areas where they may be struggling.

Why SISI is Relevant to Sign & Spell VR:

The SISI framework emphasizes active learning and scaffolded instruction, both of which are vital for effective language acquisition. By focusing on clear, structured learning with built-in feedback and opportunities for self-assessment, SISI fosters a deeper understanding of the material, especially in language learning. In the context of Sign & Spell VR, this methodology is implemented in an immersive and interactive VR setting, making the learning process both engaging and effective.

The integration of this teaching framework within Sign & Spell VR helps ensure that the application not only provides hands-on practice with immediate feedback but also encourages a gradual release of responsibility, where learners move from highly supported learning to independent practice. The inclusion of the SISI framework in Sign & Spell VR enhances its potential to address linguistic diversity and engagement, key aspects when teaching a language like Irish Sign Language (ISL), which involves complex gestures and cultural nuances.[20]

2.5. Existing Final Year Projects

2.5.1 Sign Language Translator (SLT)

Title: Sign Language Translator (SLT)

Student: Andrei Botnari

Description (brief): This project is about creating a system that can translate Irish Sign Language (ISL) gestures into text using the Xbox Kinect Sensor. This student developed a recognition pipeline that captured depth data, segmented the hands from the background, and matched the input against predefined gestures. From this project, the aim is to evaluate the accuracy the Kinect sensor under different environmental conditions such as lighting, distance, and motion.

What is complex in this project: The real time gestures recognition using low-resolution depth data. From the project, ISL contains many hand shapes that appear visually similar, making classification and tracking difficult. Managing environmental interference and differentiating between similar gestures (e.g., E, S, and T) required careful calibration and algorithm tuning.

What technical architecture was used: The student used the Kinect V2 depth sensor connected to a PC and developed in Python and C#. This project followed the Spiral Development model, which emphasises the iterative testing of gesture recognition accuracy and usability. The architecture included modules like image capture, segmentation, and features extraction before translating gestures to text output.

Explain key strengths and weaknesses of this project, as you see it:

Strengths: Hard focus on experimental validation, detailed testing across varied conditions. The spiral model ensured improvements. It provided an early exploration into ISL recognition, which remains a niche and research area.

Weaknesses: This project depended heavy on the Kinect hardware, which is now outdated and sensitive to lighting and distance as the resolution is low. It lacked interactive feedback, which means that users could not correct or practice signs dynamically, which limits learning effectiveness. [18]

2.5.2 VR Music Learner

Title: VR Music Learner

Student: Maciej Golubski

Description (brief): This student developed a virtual reality environment where users could learn musical instruments such as guitar or piano in an immersive, game-like setting. He integrated VR Gameplay, sound recognition, and a connected mobile app to track user progress and statistics. It aimed to make music learning more engaging through interaction, rhythm matching, and score tracking.

What is complex in this project: The challenge the student had from this project is the VR motion input, audio recognition, and mobile data communication. Maintaining accurate timing between the players performance and in-game feedback required strong optimization.

What technical architecture was used: This project is developed in Unity using C#, and using the

Oculus Quest Headset for VR interaction, a web server and database for the scoring system, and a mobile companion app for data visuality. The student used Agile/Scrum workflow for iterative development and user testing.

Explain key strengths and weaknesses of this project, as you see it:

Strengths: Good integration with VR, mobile, backend technology. The focus is strongly seen on gamifying it and feedback loops that enhance learning and motivation. This shows well-thought user experience for the user.

Weaknesses: It has limited educational depth. Mainly focuses on music theory. System performance may change depending on network and the accuracy of audio input. The student could have designed it from a broader perspective of testing and adding additional learning elements. [17]

2.6. Conclusions

The research conducted in this chapter reveals that there is a broad spectrum of existing solutions for sign language learning, each varying in technological approach, interactivity, and educational methods. While tools like ASL Fingerspeller, Silent Classroom VR, and ASL Champ! demonstrate the potential for VR-based sign language learning, none fully address the specific needs of Irish Sign Language (ISL) learners. The analysis of these tools, framed within the FURPS model, highlighted important lessons that directly informed the design and functionality of *Sign & Spell VR*. Notably, the importance of incorporating challenge elements, such as dynamic quizzes and personalized learning progression, was emphasized.

However, limitations in existing solutions, such as the lack of real-time feedback, poor adaptation for beginners, and limited language focus, served as key drivers for the design of *Sign & Spell VR*. By offering a more interactive experience and tailoring the application to ISL learners, the project aims to fill the gaps identified in the literature. The review also highlighted the importance of keeping the system simple and intuitive for users who may have no prior experience with VR or sign language. In contrast to more immersive classroom-based applications, *Sign & Spell VR* prioritizes short, focused practice loops, enabling users to learn in bite-sized sessions.

The research also delved into the technologies and frameworks used to build *Sign & Spell VR*. By opting for Godot 4 and leveraging the Meta Quest 3's hand tracking capabilities, the project avoids the complexities associated with other engines like Unity or Unreal, making it more accessible for a student-level VR prototype. Technologies such as GDScript, OpenXR, and Blender were carefully selected to streamline development, reduce overhead, and create an immersive, stable experience. Moreover, the decision to rely on file-based storage for user progress, rather than complex database solutions, reflects the project's aim to balance functionality with simplicity.

On the ethical side, the review stressed the importance of cultural sensitivity and the need for community involvement in the development process. Research on the FATE landscape, sign language recognition, and AI fairness further emphasized the responsibility of developers to ensure that sign language applications are not only technically proficient but also culturally respectful. For *Sign & Spell VR*, this means avoiding the use of AI-driven sign recognition systems and instead opting for a rule-based system that prioritizes accuracy and user clarity. Moreover, the integration of the SISI (Strategic and Interactive Signing Instruction) framework in the project ensures that learning is scaffolded, interactive, and accessible, aligning with best practices for language acquisition.

In conclusion, this chapter has not only reviewed and critiqued existing solutions but also integrated these insights into the design of *Sign & Spell VR*. By carefully considering existing research,

technological capabilities, and ethical standards, the project has laid a solid foundation for further development, ensuring that the final product is both effective in teaching ISL and respectful to the Deaf community. The insights gained here will guide future iterations of the system, with the potential for expanding the scope beyond the ISL alphabet to more complex vocabulary and broader educational features.

3. System Analysis

3.1 System Overview

3.1.1 What is Sign & Spell VR?

Sign & Spell VR is a virtual reality application developed to support the learning of the Irish Sign Language (ISL) alphabet in a more interactive way. The system is mainly aimed at beginners, such as students or individuals with little to no prior experience in ISL, especially those learning independently without access to formal classes.

At the start, the project aimed to include a wider range of features, including learning words and simple sentence structures. However, as development progressed, it became clear that this was not realistic within the time available. In particular, the difficulty of achieving reliable hand tracking and gesture recognition made it harder to expand the system beyond basic signs. Because of this, the scope was reduced to focus on the ISL alphabet.

While this might seem like a limitation, it allowed more time to focus on making the core interaction work properly. In that sense, the system became more of a focused prototype rather than a full learning platform, which better reflects what could realistically be achieved.

The main goal of the application is to make sign language learning more engaging compared to traditional methods such as books or videos. By placing the user in a VR environment where they can actively practise hand movements, the system attempts to create a more hands-on learning experience.

3.1.2 How the System Works

When the user puts on the Meta Quest 3 headset, they are placed into a simple virtual environment designed to reduce distractions. The system uses the headset's built-in sensors to track the position and movement of the user's hands in real time, which are then represented as virtual hands within the scene.

The application is divided into sections, allowing users to first learn about ISL, then practise individual letters, and finally attempt a basic test. During practice, the system compares the user's hand pose against predefined ISL letter shapes to determine whether the gesture is correct.

Although this process is intended to feel smooth, in practice it was not always fully reliable. Hand tracking sometimes struggled with certain letters, especially those with similar shapes. This meant that recognition results could be inconsistent, which then affected the feedback given to the user. Because of this, adjustments had to be made to keep the feedback simple and avoid confusing the user further.

3.1.3 User Experience and Interface

The user interface was designed with simplicity as a priority, mainly because the target users may not be familiar with VR or sign language. The main menu includes only a small number of options, such as "ISL Info", "Start Learning", and "Test Yourself", to avoid overwhelming the user.

During early development, there was an attempt to include more features and options on screen. However, this quickly became difficult to manage in VR, where too many elements can make interaction unclear. As a result, the interface was simplified by using larger buttons, fewer choices, and clearer visual cues.

Navigation is handled through basic hand interactions rather than more complex gestures. While more advanced controls were considered, they were often unreliable or unintuitive during testing. Keeping interactions simple made the system easier to use, even if it limited the level of control available to the user.

3.1.4 Practice and Feedback

The core of the system is based on a practice and feedback loop. Users are shown a letter and asked to replicate the correct hand shape. A 3D model is displayed to guide the user during this process.

When a sign is recognised correctly, the system provides simple feedback, such as confirming the letter. If the sign is incorrect, the system prompts the user to try again or suggests what the system detected instead.

Although this works at a basic level, the feedback is not always fully accurate. This is mainly due to limitations in hand tracking and the use of rule-based recognition. In some cases, similar signs may be misidentified. Because of this, feedback was kept simple rather than overly detailed, as more complex feedback could become misleading if the detection itself was not reliable.

Even with these limitations, the repeated cycle of attempting, receiving feedback, and retrying still supports learning over time.

3.1.5 Inspiration and Aims

The design of Sign & Spell VR was influenced by both the original project idea and findings from the literature review. Existing VR learning applications highlighted the importance of interactivity, while research suggested that immediate feedback and simple interfaces are important for beginner learners.

Rather than trying to replicate more complex systems, the project shifted towards creating a smaller, more focused experience. This was partly due to technical challenges, but also because a simpler system was more achievable within the given timeframe.

In its current form, the system reflects a balance between the original goals and the practical challenges faced during development. While it does not fully achieve everything that was initially planned, it demonstrates a working approach to using VR for sign language learning and provides a starting point for future improvements.

3.2 Requirements Gathering

3.2.1 Stakeholders?

For this project, it was necessary to identify the main stakeholders. This helped decide which features were important and which ones could be left out. At the beginning, the focus was mostly on

the technical side, but over time it became clear that different users would expect different things from the system.

The project considered both direct users of the application and others who might use it or be affected by it. Looking at these groups made it easier to make decisions about usability, feedback, and how complex the system should be. Instead of trying to include too many features, the stakeholder analysis helped keep the system more focused and suitable for its main users.

3.2.1.1 New Learners

New learners form the primary target group for the system. These are users who are not only unfamiliar with Irish Sign Language, but in many cases are also inexperienced with virtual reality as a medium. Because of this, the main challenge is not just teaching ISL but doing so in a way that does not overwhelm the user from the start.

Persona: Sophie

Sophie is a college student who is interested in learning ISL but has no prior experience. She is also new to VR, which means that both the learning content and the technology itself can feel slightly intimidating at first. From her perspective, even simple interactions may not feel obvious straight away. As a result, she would expect the application to provide clear instructions, simple navigation, and immediate feedback when mistakes are made, so that she can understand what went wrong without feeling stuck.

During development, it became clear that designing for users like Sophie required reducing complexity more than initially expected. Earlier versions of the interface included a wider range of options and interactions, which seemed useful at first, but in practice made the system harder to follow. These were later simplified after recognising that too many features made the experience confusing rather than helpful. While this does limit some advanced functionality, it creates a more approachable starting point, which is more suitable for beginner users.

3.2.1.2 Teacher and Instructors

Teachers and instructors represent a secondary stakeholder group. While the system is not fully designed for classroom management, it was considered that educators may use it as a supplementary learning tool.

Persona: Martin

Martin is an ISL instructor who teaches adult learners. He values tools that are straightforward to set up and do not require a high level of technical knowledge. His main concern is whether the application can support independent practice and reinforce what students have already learned in class, rather than replacing traditional teaching.

When considering Martin's needs, some limitations of the current system became more noticeable. Features such as progress tracking, structured lesson monitoring, and multi-user management were considered during development, but were not fully implemented due to time constraints. This means that while the system can still be useful as an additional learning aid, it may not fully support classroom-based use. This highlights a gap that could be addressed in future development if the system was expanded further.

3.2.1.3 Deaf Community

The Deaf community (ISL users, interpreters, advocates) represents an important stakeholder group, particularly in ensuring that the system is respectful and accurate in its representation of ISL. This includes interpreters, fluent ISL users, and advocates.

Persona: Aisling

Aisling is an ISL interpreter who supports the development of tools that improve accessibility but is cautious about how sign language is represented digitally. She is especially concerned about incorrect or oversimplified signs, as these could mislead learners or reduce trust in the system. From her perspective, learning tools should aim to reflect both the linguistic and cultural aspects of ISL as accurately as possible.

This raised a challenge during development, as the system currently relies on rule-based gesture recognition rather than expert validated or data driven models. While this approach works for a prototype, it does introduce the risk that some signs may not be interpreted fully accurately, especially those with subtle differences. This does not make the system unusable, but it does suggest that further validation with ISL users or experts would be necessary in future versions. It also reinforces that the current system should be viewed as a learning aid rather than a fully reliable reference for ISL.

3.2.1.4 Potential Stakeholders

Potential stakeholders include groups who may benefit from the system in the future, such as families, schools, or researchers. Although they were not the focus during development, they were still considered to ensure that the system design does not become too limited or difficult to expand later.

Persona: Mrs. O'Connor

Mrs. O'Connor is a parent of a Deaf child and is interested in tools that allow families to learn ISL together. She values accessibility, ease of use, and the ability to revisit lessons at her own pace. For her, the system needs to feel flexible rather than structured or restrictive.

While considering users like Mrs. O'Connor, features such as cross-device support and shared learning were explored. However, these were not implemented in the current version, mainly due to the need to focus on core functionality such as hand tracking and feedback. This reflects a broader design decision within the project, where priority was given to building a stable and working prototype before expanding into more advanced features. Although this limits the system in its current state, it keeps the development process more manageable and provides a clearer foundation for future improvements.

3.2.2 How Requirements were Identified

To decide what features the ISL VR application should include, a mix of research, informal feedback, and trial-and-error during development was used. Existing sign language learning apps and some academic papers were looked at to get a general idea of what works well, especially in VR environments. At the same time, user reviews and forum discussions were also checked, as these tend to show what people struggle with when using similar apps.

Some informal conversations were also carried out with classmates and friends who were interested in learning sign language, along with a few people familiar with ISL. These weren't formal interviews

or anything like that, but they still gave a rough idea of what users might expect. That said, the feedback is quite limited and not representative of a wider group. Because of this, a plan for later stages is to use simple questionnaires once the prototype is more stable, so that more structured feedback can be collected.

Experience from the XR Prototyping course also played a role. During development, small notes were made when something didn't feel right, or when a certain interaction worked better than expected. These observations were then used to adjust the requirements over time. This approach is quite informal, but it fits with how the project developed, where ideas changed as the system was being built rather than being fully planned from the start.

3.2.2 Initial Requirements List

Based on the research, feedback, and early testing with the prototype, a set of initial requirements was put together to guide the development of Sign & Spell VR. These mainly focus on making the system easy to use and stable, rather than trying to include too many advanced features.

- The app should use Meta Quest 3 hand tracking so users don't need controllers. This was chosen to make the experience feel more natural, although it can sometimes be less reliable.
- The system should be easy to follow, so users can get consistent results even if their hand position or size is slightly different. This is important for beginners, but also difficult to get right.
- The interface should be simple, with clear instructions and large visual elements. While this limits how much can be shown at once, it makes the system easier to understand.
- The app should give immediate feedback when a user performs a sign, either showing that it is correct or indicating that something needs to be adjusted. This was seen as important for learning, as users need to know what they are doing wrong straight away.
- Progress should be saved so users can continue where they left off. In the current version this is quite basic, but it still improves the overall experience.
- The system should run smoothly on the Meta Quest 3, without crashing or slowing down too much. This means some design choices had to be kept simple.
- The app should be built using Godot 4, GDScript, and Blender, mainly because they are easier to work with. However, this might limit performance compared to other engines.
- Accessibility should be considered where possible, including support for beginners and left-handed users. Not all of this is fully implemented yet, but it was still kept in mind during development.

Overall, these requirements show that the focus of the project was to build something that works reliably and is easy to use, even if that meant leaving out more advanced features. Given the limited time and scope, this seemed like a more realistic approach.

3.3 Requirements Analysis

3.3.1 Organising the Requirements

After the requirements were gathered, they were organised in a way that would keep the overall scope clear and manageable. The requirements were split into two main categories: functional and

non-functional. Functional requirements focus on what the system actually does, such as recognising signs, providing feedback, or saving user progress. Non-functional requirements are more about how the system behaves overall, including aspects like performance, usability, and simplicity.

For example, a requirement like “the system must let users save progress” is considered functional because it describes a specific feature. On the other hand, something like “the system should be easy for beginners to use” is non-functional, since it relates more to the general experience rather than a single action. Keeping this separation made it easier to understand what needed to be built versus what needed to be improved or refined.

ID	Category	Requirement Statement	Source
F1	Functional	System must recognise ISL fingerspelling letters	[user interviews]
F2	Functional	System must provide immediate feedback on signs	[user interviews]
F3	Functional	Users can enter and exit a section	[user feedback]
F4	Functional	System must display 3D hand model references	[existing apps survey]
NF1	Non-Functional	App must run smoothly at 72 FPS on Quest 3	[Meta Quest specs]
NF2	Non-Functional	Interface must be intuitive for beginners	[user interviews]
NF3	Non-Functional	Hand tracking accuracy >90% for each letter	[hand tracking research]

At the same time, these requirements were not treated as fixed. As development progressed, some requirements had to be adjusted slightly, especially when certain features were more difficult to implement than expected. This made the process feel more iterative, rather than something that was fully decided at the beginning.

3.3.2 Prioritising Features

With several possible features identified, it became necessary to decide what should actually be implemented within the time available. The focus was placed on features that directly affect whether the system works or not, rather than those that simply improve the experience.

The main priorities were reliable hand tracking, accurate recognition of ISL signs, clear real-time feedback, and the ability to save and load user progress. Without these, the application would not function as a learning tool in any meaningful way. Features such as more advanced lesson modes, improved menus, or cloud syncing were considered, but were pushed back as they were not essential to the core experience.

To help stay on track:

- **Must-have:** Core hand tracking, sign recognition, feedback, basic navigation, and saving progress.
- **Should-have:** Additional lesson types, improved accessibility, possible progress tracking or charts.

- **Nice-to-have:** Visual improvements, custom avatars, advanced quizzes, and analytics.

This prioritisation reflects a practical decision rather than an ideal one. In a more complete system, many of the “should-have” features would likely be included. However, given the limited scope of the project, focusing on a smaller number of stable features was seen as more realistic. In a way, this meant accepting that the system would be somewhat limited, but also more reliable.

3.3.4 Making an Initial System Model Use Case

A use case diagram was created to show how a typical user would interact with the system. The main actor is the learner, representing someone using the VR headset to practise ISL. The diagram includes actions such as learning letters, practising signs, taking tests, and viewing results.

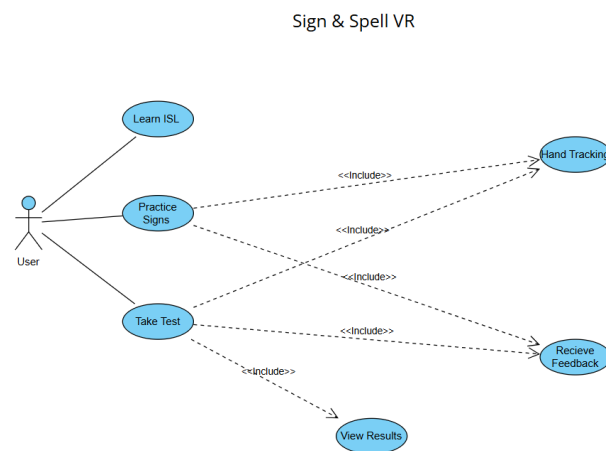


Figure: Use Case Diagram for Sign and Spell VR

Looking at the diagram, it becomes clear that a lot of these actions depend on the same underlying processes. For example, both practising and testing rely on hand tracking and gesture recognition, which means these parts of the system need to be reliable before anything else works properly. Similarly, features like feedback and saving progress are reused across different actions, rather than being separate systems.

This helped highlight an important point during development. Even though the system appears to have several features on the surface, many of them are built on a small number of core functions. Because of this, more effort was placed on getting those core parts working first, rather than trying to build too many separate features at once. This decision helped keep the system more stable, although it did limit how much functionality could be added overall.

3.3.5 Risk and Challenges

Several risks and challenges were identified during development, both technical and usability related. One of the main technical challenges is the accuracy of hand tracking. VR hand tracking is not always consistent, and certain ISL letters can look very similar, which increases the chance of incorrect recognition. If the system fails to recognise signs correctly, it could quickly lead to frustration for the user.

Performance is another concern. If the system becomes too complex, especially with real-time feedback and tracking running at the same time, it may cause lag or even crashes on the headset. This creates a trade-off between adding more features and maintaining stable performance.

There is also a usability risk, particularly for users who are new to VR. Even basic interactions may feel confusing at first, which could make the learning experience less effective. Because of this, keeping the interface simple became more important than adding extra functionality.

To reduce these risks, the project focuses on simple interaction flows and basic calibration where possible. Early testing with a small group of users is also planned, so that major issues can be identified before the system becomes too complex to change easily. While this does not remove all risks, it helps reduce the likelihood of major problems later.

3.4 Logical Architecture of System

This section brings together the main requirements of the system into a clearer structure that can guide development. It outlines what the system is expected to do, how it should behave, and how the main components fit together. At this stage, the architecture is still quite simple, as the focus is on building a working prototype rather than a fully developed system.

3.4.1 Initial Functional Requirements

These requirements describe the main features that the application needs in order to function as a basic ISL learning tool:

- The system must track both hands and all five fingers using the Meta Quest 3 headset. This is necessary so users can practise correct hand shapes, although in practice the tracking may not always be fully accurate.
- The system should recognise ISL signs and provide immediate feedback to the user. This allows learners to understand whether they performed a sign correctly, but depends heavily on how reliable the recognition system is.
- The application should display 3D hand models (created in Blender) to show users what the correct sign looks like. This helps guide learning, although it adds some extra performance overhead.
- The system should include a simple navigation system, with menus for lessons, practice, and feedback. This was kept minimal to avoid confusing users who are new to VR.
- Users should be able to continue from where they left off, so that progress is not lost between sessions. In the current version, this is implemented in a basic way.
- The application should include at least one practice mode (such as the alphabet), with some way of tracking basic progress. More advanced features were considered but not included at this stage.

Overall, these functional requirements focus on making the system usable and interactive, rather than trying to include too many features at once.

3.4.2 Initial Non-Functional Requirements

Non-functional requirements describe how the system should behave and the overall quality of the experience:

- The application should be easy to use, even for users with no prior experience of VR or ISL. This means keeping menus simple and instructions clear, although this sometimes limits how much information can be shown.
- The system should run smoothly on the Meta Quest 3. Lag or crashes would likely make the experience frustrating, so performance had to be considered when adding features.
- Accessibility should be considered where possible. This includes support for left-handed users and making text large enough to read comfortably in VR. Not all of this is fully implemented yet, but it was kept in mind during development.
- The system should be reasonably easy to maintain and extend in the future. For example, adding new signs or features should not require major changes to the whole system, although this has not been fully tested yet.

These requirements show that usability and stability were prioritised over more advanced or complex features.

3.4.3 Initial System Architecture

Having a basic structure for how the system components interact helps make development more manageable. At this stage, the architecture is kept fairly simple and is mainly split into three core modules:

Main Components/Modules:

1. Hand Tracking Module:

- Uses the Meta Quest 3's built-in tracking (via OpenXR) to capture hand and finger positions in real time.
- Sends this data to the feedback system for processing.
- This is one of the most important parts of the system, but also one of the least predictable in terms of accuracy.

2. Feedback Engine:

- Compares the user's hand position against predefined ISL signs.
- Determines whether the sign is correct or not.
- Provides immediate feedback such as "Correct" or indicates that the user should try again.
- This part of the system is relatively simple in the current version and may not handle more complex signs well.

3. User Interface (UI) Module:

- Handles menus, lesson selection, and feedback display.
- Also displays 3D hand models to demonstrate signs.
- Designed using Godot 4 with large, simple UI elements to suit VR interaction.

Although this architecture is quite basic, it reflects the current scope of the project. More advanced systems, such as detailed analytics or adaptive learning, were not included at this stage.

3.4.4 How it all works

The overall flow of the system can be described in a few steps:

- The user puts on the headset and selects a mode from the user interface.
- The Hand Tracking Module monitors the user's hand movements in real time.
- This data is passed to the Feedback Engine, which checks it against stored ISL gestures.
- The system then returns feedback, indicating whether the sign was correct or needs adjustment.
- The UI displays this feedback along with any additional information, such as progress or messages.

This flow shows that most of the system relies on the interaction between tracking and feedback. If either of these parts does not work well, the overall experience is affected. Because of this, more focus was placed on getting these core interactions working reliably rather than expanding the system further.

3.5 Conclusion

Overall, this chapter outlined the main analysis and design decisions behind Sign & Spell VR. It covered how the system was defined, who it is intended for, and what requirements were considered important for development.

One of the main points that became clear throughout this chapter is that the system had to be simplified compared to the original idea. While the initial plan included more advanced features, such as word-level learning and more detailed feedback, these were not realistic within the available time. Instead, the focus shifted towards building a smaller but more stable system based around ISL alphabet learning.

The requirements and architecture reflect this decision. More emphasis was placed on core features such as hand tracking, feedback, and usability, rather than trying to include too many additional features. This helped keep the system manageable, although it does mean that the current version is somewhat limited in scope.

At the same time, the analysis also highlighted a few challenges, particularly around hand tracking accuracy and overall usability. These issues influenced many of the design decisions, such as keeping the interface simple and limiting the complexity of feedback.

Overall, this chapter provides a clear foundation for the next stage of the project. The system design is not fully complete, but it is structured enough to guide implementation and further development.

4. System / Experiment Design

4.1 Introduction

4.1.1 Translating Requirements into Design

The purpose of this section is to explain how the requirements identified in the previous chapter are translated into a working system design for Sign & Spell VR. This involves deciding how the system will be structured, how different components will interact, and which technologies are most suitable for implementing the required features.

Rather than treating the requirements as fixed, they were used as a guide to shape design decisions throughout development. In some cases, this meant simplifying or adjusting features so that they could be realistically implemented within the given time. Because of this, the design reflects a balance between what was originally planned and what could actually be achieved in practice.

4.1.2 Design Principles and Goals

The main design principles used in this project are simplicity, modularity, and user-friendliness. These were chosen based on the needs of beginner users and the limitations identified earlier.

Simplicity was important to ensure that users could understand how to use the system without needing prior experience in VR. This influenced decisions such as reducing menu complexity and limiting the number of on-screen elements.

Modularity was used to separate different parts of the system, such as hand tracking, feedback, and user interface. This made it easier to test and adjust individual components without affecting the entire system, although this structure is still quite basic in the current version.

User-friendliness was also a key goal, particularly in terms of navigation and feedback. Clear instructions and simple responses were prioritised over more detailed or complex features. While this improves accessibility, it also limits how much information can be given to the user at once.

4.2. Software Methodology

4.2.1 Incremental Model

The Incremental Model was chosen as the development methodology for this project. Instead of attempting to build the entire system at once, the work was divided into smaller stages, where each stage added a working part of the application.

The first increment focused on the core functionality, including hand tracking on the Meta Quest 3, basic ISL letter recognition, and simple feedback such as “correct” or “try again.” These features were prioritised because the system would not function as a learning tool without them.

Later increments introduced additional features such as menu navigation, progress saving, and the display of 3D hand models. By building the system step by step, it was possible to test each part before adding more complexity. This reduced the risk of major issues appearing later in development.

Although this approach worked well overall, some features still required reworking as new challenges appeared, particularly with hand tracking accuracy. This shows that even with an incremental approach, some level of iteration is still necessary.

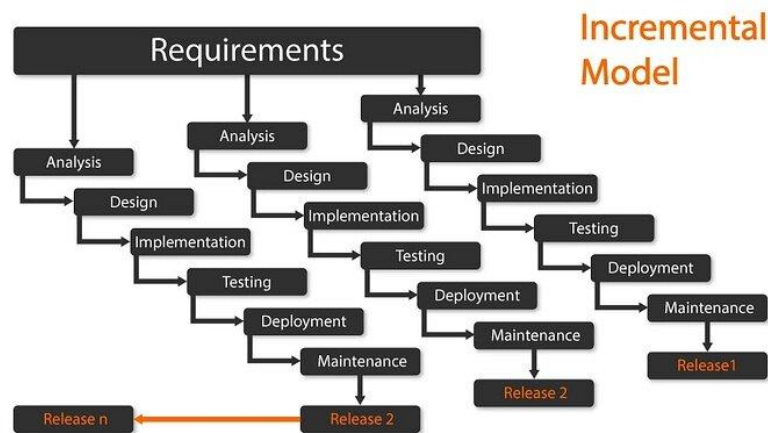


Figure 5: Incremental Model

4.2.2 Advantages of Incremental Development for my Project

The incremental approach was particularly suitable for this project, as it allowed steady progress while keeping the system functional at each stage. Having a working version early on made it easier to identify issues, especially with hand tracking and user interaction.

It also helped ensure that the most important features were completed first. Instead of focusing on less important additions, the development stayed centred on core functionality such as sign recognition and feedback. This reduced the risk of ending up with an incomplete system.

Another advantage was flexibility. If a feature proved too difficult or time-consuming, it could be delayed or simplified without affecting the rest of the system too heavily. This was useful given the technical challenges involved in VR development.

However, this approach also meant that some features were left at a basic level, as time had to be managed carefully across each increment. As a result, the system is functional but not fully developed in all areas.

4.2.3 Application Approach

In practice, development started with a basic prototype focusing on essential features. This included real-time hand tracking, initial ISL letter recognition, and simple feedback. The main goal at this stage was stability, as unreliable tracking would affect all other features.

Once this base was working reasonably well, additional features were added in later stages. These included the full alphabet, improved menu navigation, progress saving, and 3D hand models for guidance. Each stage was tested on the actual headset before moving on, which helped identify issues early.

This step-by-step approach made development more manageable, although it did slow down the addition of more advanced features. By the end, the system reached a point where it could function as a complete prototype, even if some areas still require improvement.

4.3 Project Planning

This section provides an overview of the approach taken to ensure that the Sign & Spell VR project was developed within the given timeframe and resources. Effective planning was crucial for maintaining a clear structure throughout the development process and ensuring that each milestone was achieved on time. This section covers the development approach, project schedule, and planning constraints that shaped the overall timeline.

4.3.1 Development Planning Approach

The development of *Sign & Spell VR* followed a structured yet flexible planning approach. The primary focus was on delivering core features incrementally, as discussed earlier with the Incremental Model. The development was divided into phases to ensure that each component—such as hand tracking, gesture recognition, feedback system, and user interface—was developed, tested, and refined before moving on to the next phase. This allowed the project to stay on track and ensured that foundational components were solid before adding complexity.

The key stages in the development process included:

1. **Prototype Development:** The first phase focused on creating a basic prototype that handled the most essential features, such as hand tracking and basic ISL letter recognition. This prototype was designed to be as simple as possible, ensuring that the core functionality was working before expanding the system.
2. **Feature Expansion:** Once the prototype was stable, the next phase involved adding new features such as advanced gesture recognition, real-time feedback, and user interface enhancements. Each of these features was developed in stages, allowing for regular testing and feedback from users to improve the system incrementally.
3. **Testing and Refinement:** After the initial development phases, continuous user testing was incorporated into the development process. Each feature was tested for usability, performance, and educational effectiveness. Based on the feedback received, the system was refined, with bug fixes and performance improvements made regularly.

This iterative development approach allowed for consistent progress while ensuring that important features were prioritized early on. It also meant that the development process remained adaptable, with adjustments being made as issues arose during testing and user feedback.

4.3.2 Project Schedule

The Project Schedule was created to ensure that development proceeded according to plan, with all critical milestones being met on time. The schedule was designed to balance the time required for developing core features, testing, and refining the system. Since this project was being developed individually, it was essential to manage time effectively to avoid any major delays.

Key elements of the project schedule included:

- **Core Development Period:** The initial focus was on hand tracking and gesture recognition features. This phase took approximately 6 weeks, with additional time set aside for integration with the UI Module and the Feedback Engine.
- **User Testing and Refinement:** Once a main feature was complete, user testing began. During this phase, the system was tested for usability and performance. Testing and feedback loops accounted for allowing for iterations and adjustments during the core development period.
- **Final System Testing and Polish:** The final phase of the project focused on polishing the system and ensuring that all components worked seamlessly together. This phase included final testing and bug fixes, which took 2 weeks to complete.

By structuring the schedule around key milestones, it ensured that there was enough time to complete each section while still leaving room for adjustments based on feedback and unforeseen challenges.

4.3.3 Gantt Chart

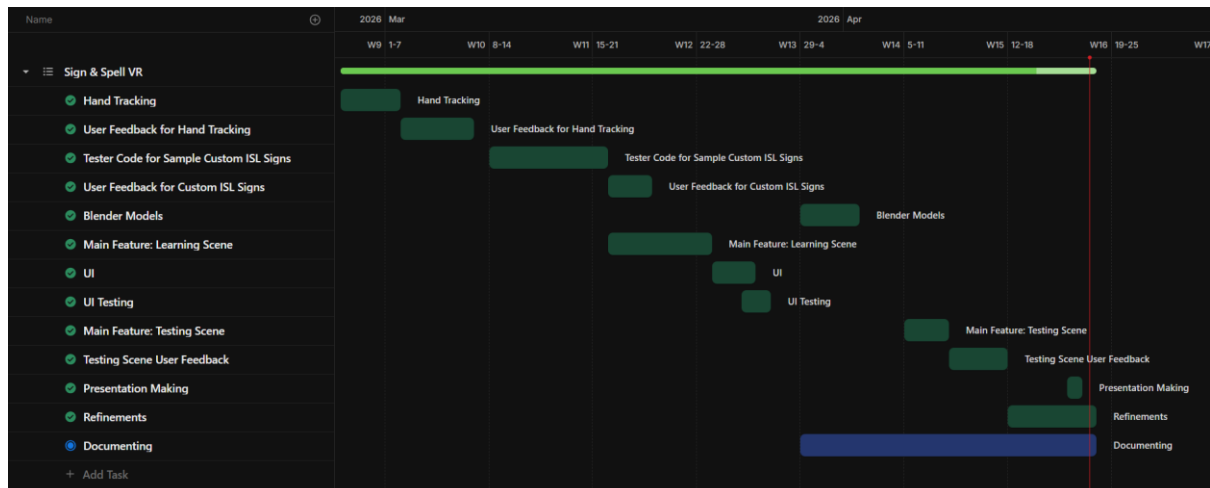


Figure 6: Gantt Chart

The Gantt chart presented above outlines the timeline and task management for the Sign & Spell VR project from March 2026 to April 2026. The chart provides a visual representation of the key phases and tasks, ensuring that the project progressed in a structured and organized manner. Each task was carefully planned and allotted a specific time frame, and dependencies between tasks were considered to ensure a logical flow.

1. **Early Development (W1 - W4):**
The first phase of the project focused on setting up the core features, such as hand tracking and gathering user feedback for hand tracking. These tasks were crucial to establish the primary interaction method for the app. Simultaneously, work on the custom ISL signs and Blender models started.
2. **Interface and Scene Development (W4 - W6):**
After establishing the core functionality, the focus shifted to building the Learning Scene and the User Interface (UI). The UI was tested during this phase, and user feedback was collected

for custom ISL signs. This phase ensured that the core learning features were ready for testing by users.

3. Testing and Refinements (W6 - W8):

The app moved to the testing phase, with the Testing Scene developed and user feedback gathered to refine the system. Refinements were made based on the collected feedback to improve the accuracy of hand tracking and feedback responsiveness. During this period, the team also focused on preparing the presentation.

4. Finalizing and Documenting (W8 - W9):

The final phase of the project involved making final refinements, testing the system thoroughly, and preparing the documentation and presentation. This stage ensured the project was ready for final evaluation and review.

4.3.4 Planning Constraints and Adjustments

While the project planning was designed to keep the development process on track, there were several constraints and adjustments made during the project. These were primarily due to the limitations of resources, time constraints, and challenges encountered during development. Some of the main constraints and adjustments were:

- **Time Constraints:** Given the limited time available for development, certain features had to be deprioritized or simplified. For example, advanced gesture recognition and the inclusion of more complex ISL signs were initially planned but had to be delayed due to time limitations. As a result, the system focused on the basic ISL alphabet rather than expanding to include full sentences or more complex vocabulary.
- **Technical Challenges:** One of the significant technical challenges encountered was the accuracy of hand tracking. While the Meta Quest 3 provides robust hand tracking capabilities, it was found that lighting conditions and user hand positioning affected recognition accuracy. To address this, extra time was spent fine-tuning the system's fitness function and gesture recognition algorithms to handle varying user conditions.
- **Simplified UI:** Initially, the UI Module was intended to be more complex, incorporating additional interactive elements such as progress charts and detailed instructional screens. However, given the project's time constraints, the UI was simplified to focus on the basics: navigation, feedback, and basic instructions. This was an adjustment made to ensure that the core functionality was fully developed before adding additional features.
- **Resource Limitations:** As the project was developed individually, there were limitations on available resources, particularly in terms of time and expertise. Some features that would require extensive development (such as cloud syncing or multi-user support) were not included in the project's scope due to time constraints. These features were deferred for future iterations.

Despite these constraints, the planning approach allowed the development to stay on track, with adjustments being made as necessary. The focus on core features and user feedback ensured that the system remained functional and effective for its primary goal: helping users learn ISL in a VR environment.

4.4 Overview of System

This section breaks down how the different parts of the Sign & Spell VR system work together cohesively. The aim is to provide an understanding of how the system is structured both logically and physically, and how data flows between the components. Additionally, we will discuss the design choices made during the development process, particularly the decision to implement a custom-designed hand gesture recognition system using the Hand Pose Detector library.

The system is built around three core components: hand tracking and gesture recognition, feedback engine, and the user interface (UI) module. These components were carefully selected and designed to work in tandem, ensuring that the system not only meets functional requirements but also provides a smooth and intuitive learning experience for users.

4.4.1 Logical Architecture

The logical architecture of the system represents the conceptual design and structure of the software components, highlighting how they interact to fulfil the user's educational needs. The logical architecture is divided into several distinct modules, each responsible for a specific task in the overall system. Below is an overview of these modules:

1. Hand Tracking and Gesture Recognition Module:

- This module is the heart of the Sign & Spell VR system. It leverages the Meta Quest 3's built-in hand tracking sensors, capturing real-time data on the user's hand movements and positions. The Hand Pose Detector library is at the core of this process, analysing the captured hand movements and comparing them against custom-designed ISL gestures created specifically for the application.
- The fitness function, integrated within the Hand Pose Detector library, compares the detected hand pose to the designed ISL signs and evaluates the degree of accuracy. This function considers finger positions and hand orientation, ensuring that variations in users' sign-making are handled accurately.
- One of the key advantages of this approach is that it avoids predefined templates for gestures, allowing for more flexible sign recognition. The system is able to dynamically compare the user's hand gestures with the custom-built ISL alphabet, which has been specifically calibrated to align with the unique characteristics of ISL signs.

2. Feedback Engine:

- The Feedback Engine is responsible for providing immediate responses based on the user's input. After the hand gesture is captured and processed, the engine assesses whether the user's sign is accurate by evaluating the output from the fitness function.
- If the hand pose matches the desired ISL sign, positive feedback (e.g., "Correct!") is displayed to the user. If the sign is incorrect or deviates from the expected gesture, corrective feedback (e.g., "Try again") is displayed to encourage improvement.
- The feedback engine operates in real time, ensuring that users receive immediate responses, which is critical for effective learning. The system also takes into account

minor variances in sign execution, providing a degree of flexibility while maintaining accuracy.

3. User Interface (UI) Module:

- The UI Module is designed to provide users with clear visual cues and feedback. It includes navigation elements for transitioning between lessons, and it displays the hand signs that the user is attempting to learn.
- 3D hand models are used to demonstrate the correct hand position for each ISL gesture, offering users visual guidance during their learning process. The UI also includes buttons for navigating between lessons and progress tracking elements to show the user how far they've come.
- The UI is kept simple and minimalistic to avoid overwhelming the user, especially beginners who may not be familiar with VR environments. The goal is to create a clean, intuitive interface that allows users to focus on learning ISL without being distracted by unnecessary complexity.

The logical architecture focuses on providing a cohesive, user-centered learning experience. The seamless integration of hand tracking, gesture recognition, feedback, and user interface is central to the system's functionality.

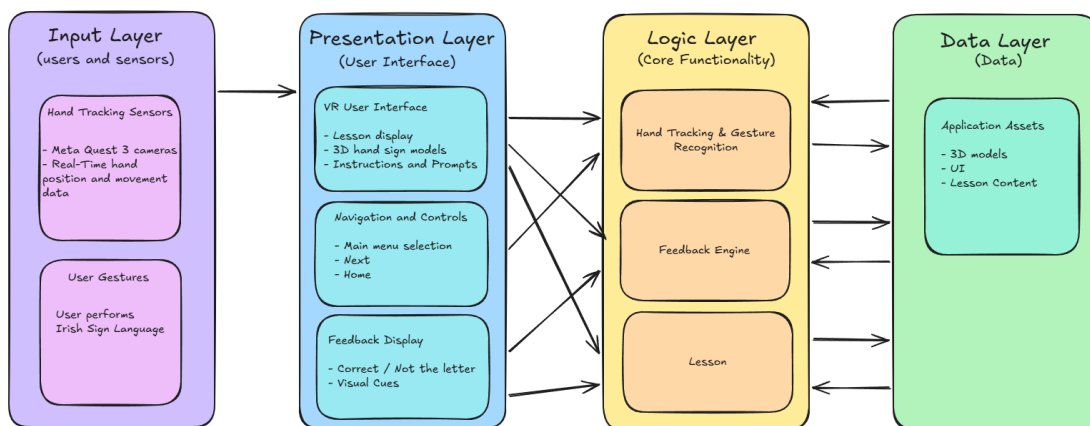


Figure 7: Logical Architecture Diagram

4.4.2 Main System Module

The Main System Module serves as the control centre for *Sign & Spell VR*. It is responsible for orchestrating the communication between the various components of the system, ensuring that all parts of the system are working together in harmony.

- **System Coordination:** The Main System Module manages the flow of data between the Hand Tracking Module, Gesture Recognition System, Feedback Engine, and UI Module. When the user performs a gesture, the hand tracking data is captured and passed to the Gesture Recognition System, which processes the data using the Hand Pose Detector library. The recognition results are then sent to the Feedback Engine, which updates the UI with

appropriate visual feedback. This continuous flow ensures that the system remains responsive and effective throughout the learning process.

- **User Progress and Sessions:** The Main System Module also tracks the user's progress, including which lessons they've completed and which signs they've mastered. This information is saved locally on the Meta Quest 3 using JSON files, enabling the user to return to their session later without losing any data. User sessions are managed efficiently to ensure that users can pick up from where they left off.
- **Example Scenario:** Consider a user performing an ISL sign, like the letter "A". The Hand Tracking Module detects the hand movement, sending this data to the Gesture Recognition System. The recognition system, through the fitness function in the Hand Pose Detector library, compares the hand pose to the custom-designed sign for "A". The Feedback Engine processes this information and updates the UI to display either a "Correct!" message or a "Try again" prompt. This continuous feedback loop ensures an interactive and engaging learning experience for the user.

The Main System Module is crucial for maintaining the system's synchronization, ensuring that the hand tracking, gesture recognition, and feedback processes work together smoothly and in real time.

4.4.3 Data Flow Between Modules

The data flow describes how information is passed between the different modules of the system, from hand tracking to feedback generation. Ensuring smooth data flow is crucial for providing a responsive and seamless learning experience.

1. User to Hand Tracking Module

The process begins when the user performs a hand gesture (hand input). This input is captured by the Meta Quest 3 sensors and sent to the Hand Tracking Module.

2. Hand Tracking Module to ISL Sign Data

The Hand Tracking Module analyses the position and movement of the hand and extracts the hand pose. This hand pose data is then sent to the ISL Sign Data module.

3. ISL Sign Data to Feedback Engine

The ISL Sign Data module contains the predefined dataset of ISL letters. It looks up and compares the captured hand pose with the correct ISL sign data. Once the sign is matched, the evaluated data is sent to the Feedback Engine.

4. Feedback Engine Processing

The Feedback Engine receives the hand pose data and determines whether the user's sign is correct or incorrect by comparing it with the expected ISL sign. The fitness function calculates how closely the hand pose matches the correct sign.

5. Feedback Engine to UI / User Feedback UI

Once processed, the result (correct or incorrect) is passed to the UI Module. The UI Module displays visual feedback such as “Correct!” or “Try Again,” and provides helpful hints along with 3D hand models to guide the user.

6. UI to User

The user receives the feedback, messages, and hints through the UI, helping them understand their performance and adjust their hand pose accordingly.

7. UI to Test Module

The UI also sends a progress signal to the Test Module, indicating that a sign attempt has been completed.

8. Test Module Processing

The Test Module collects performance data such as accuracy, the number of attempts, and time taken. This data is used to monitor user progress and evaluate overall system performance.

This diagram ensures that each module communicates effectively and that data flows seamlessly between them to provide real-time interaction for the user.

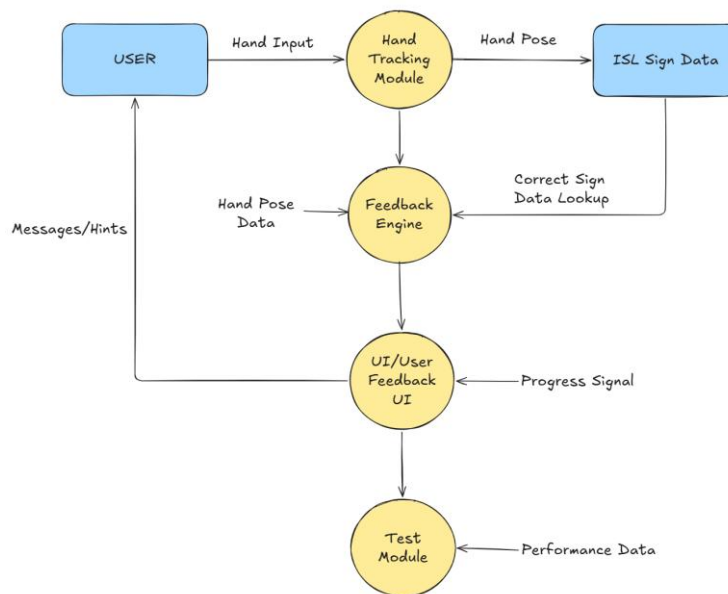


Figure 8: Data Flow Diagram

4.4.4 Modular Design Benefits

A modular design means that the system is broken down into separate, smaller components or "modules," each of which is responsible for a specific task. This design choice brings several key benefits:

- **Testability:** With a modular design, each component can be tested independently. If there's an issue with one part of the system (e.g., the feedback engine isn't showing the correct feedback), the problem can be isolated and fixed without affecting the other parts of the

system. This makes it easier to troubleshoot and ensures that the system remains reliable throughout development.

- **Flexibility and Iteration:** If one part of the system needs improvement, such as better hand tracking or enhanced feedback accuracy, it can be modified without disrupting the entire application. For instance, if the gesture recognition system needs to be updated to recognize more complex signs, the Hand Tracking Module or UI Module does not need to be altered, minimizing disruption.
- **Scalability:** A modular design also makes it easier to add new features in the future. For example, if more advanced ISL vocabulary needs to be added, this can be done by expanding the Gesture Recognition System module without changing the core structure of the app. As new technologies or learning tools emerge, they can be integrated into the existing system by updating the relevant module.
- **Simplified Debugging:** If a bug or issue arises, the modular design allows developers to trace the problem to the specific module where the error occurs. This makes debugging more efficient, as developers do not need to sift through the entire codebase but can focus on one isolated module at a time.
- **Future Proofing:** As new features or updates are introduced, they can be added in a modular fashion. If the app expands to include more advanced features like multi-user support or additional languages, these can be integrated without overhauling the system.

In summary, the modular design approach makes the system easier to build, test, maintain, and expand. It provides flexibility to adapt to changes, ensures the system remains manageable during development, and allows for future improvements as the project grows.

4.5 User Interface Design

4.5.1 Layout Designs

The layout design of Sign & Spell VR was crafted to ensure a simple, accessible, and intuitive experience for users, with a focus on making navigation via hand gestures easy to understand and use.

- **Minimalist UI:** The UI was designed to keep only the most essential elements visible to the user. Large, clear text like “Start Learning” and “Test Yourself” were prominently displayed, but navigation was mainly driven by hand gestures rather than traditional buttons or menus.
- **Navigation via Hand Signs:** Users navigated through the app using hand gestures to transition between different scenes. For example, a thumbs up gesture could be used to move to the next letter or to the testing phase. This method of interaction kept the experience seamless, allowing users to control their learning flow naturally.
- **3D Hand Models and Feedback:** In the Learning Scene, users saw 3D hand models to help them understand how to form each ISL sign. Feedback (e.g., “Correct” or “Try Again”) was displayed clearly on the screen in response to their gestures, providing real-time guidance.

- **Minimalist Controls:** Since gestures were used for navigation, the on-screen elements were minimized. Visual cues were kept simple to avoid overwhelming users, particularly beginners, by showing only key prompts like “Next” or “Back” after a gesture.
- **Ease of Use:** The layout was designed to ensure smooth transitions between actions, with gesture-based navigation reducing the need for complicated controls.

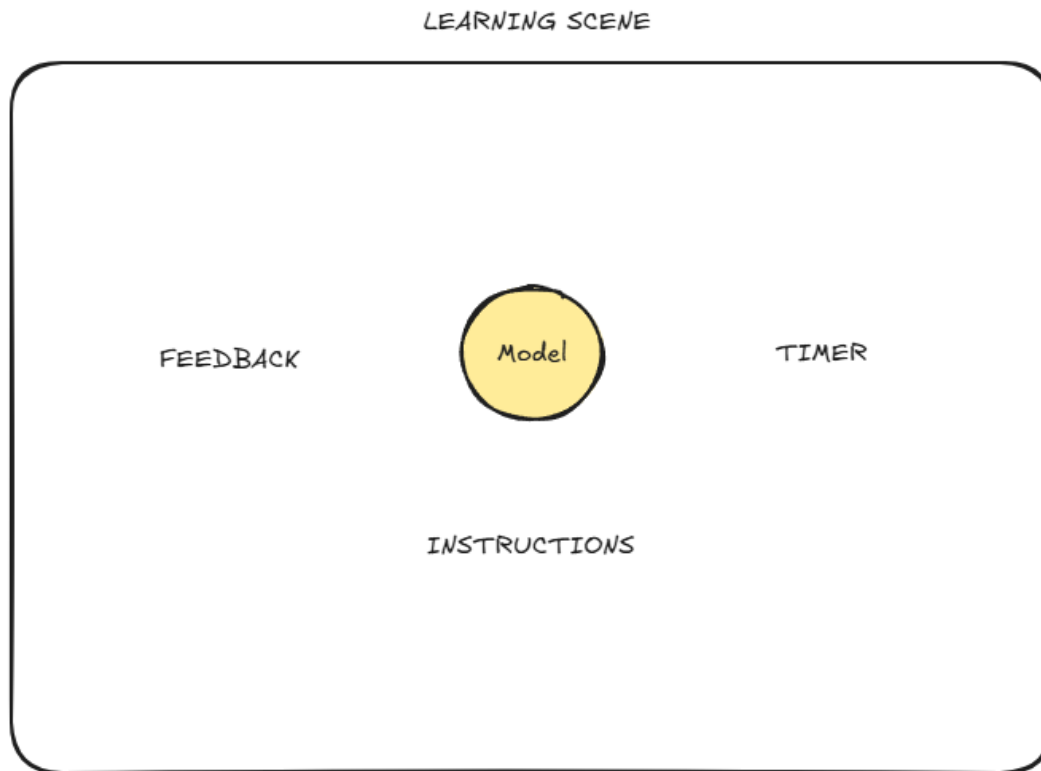


Figure 10: Sample Layout for Learning Scene (Layout)

4.5.2 Interaction Design

Interaction design focused on enabling users to navigate the app and interact with its features entirely through hand gestures, providing a natural and intuitive experience.

- **Hand Gesture Navigation:** The core interaction method was hand gestures. Users could perform gestures like a thumbs up to proceed to the next letter, or a fist to go back. Hand tracking through the Meta Quest 3 enabled real-time gesture recognition, ensuring that users could seamlessly move between learning and testing phases.
- **Real-Time Feedback:** The system provided immediate feedback to users after performing gestures. If the gesture matched the correct ISL sign, the user would see “Correct!”; if it was incorrect, they would be prompted to “Try Again”. This feedback was designed to be immediate, ensuring the user could adjust their hand position and learn from their mistakes.

- **Gesture-Based Interaction:** The system allowed users to navigate through menus and transition between scenes by performing specific hand gestures, which were recognized by the system.
- **User Flow:** The user was guided step-by-step through the learning process, from the Learning Scene (where they practiced signing ISL letters) to the Testing Scene (where they were asked to recall and sign letters). The interaction flow was designed to ensure users progressed logically, with gestures serving as both input and control throughout.

4.6 Fulfilment of System Requirements

The Fulfilment of System Requirements outlines how the Sign & Spell VR application has been developed to meet the functional and non-functional requirements. This section highlights the system's ability to perform key functions, with particular attention to how hand tracking, gesture recognition, and feedback mechanisms are integrated into the design of the ISL learning system.

Core Functional Requirements:

- **Hand Tracking:** The system uses the Meta Quest 3's sensors for real-time hand tracking. The hand movements are captured and analysed to detect the user's hand positions. The system evaluates the detected gestures by comparing them to custom-designed ISL hand signs.
- **Gesture Recognition:** The gesture recognition system compares the user's hand position to the custom ISL signs that were created for this project. It helps recognize basic ISL letters, ensuring that the user's gestures match the designed signs as closely as possible. This method ensures a good level of accuracy in recognizing basic signs of the ISL alphabet.
- **Real-Time Feedback:** After a gesture is recognized, the system immediately provides feedback. The Feedback Engine generates a response, such as "Correct!" for accurate gestures, or "Try Again" for incorrect ones. This immediate feedback is essential for encouraging learners and helping them correct their signs in real-time.
- **User Interface (UI):** The UI Module displays the feedback to the user and provides easy navigation through the lessons. It also shows 3D models of the ISL signs for users to compare their hand shapes against, helping them adjust their gestures and providing guidance during practice.

Non-Functional Requirements:

- **Performance:** The system performs well on the Meta Quest 3 headset, ensuring that the hand tracking and gesture recognition are responsive. The feedback is provided almost instantly, without noticeable delays, which helps keep the learning process smooth.
- **Usability:** The application is designed to be intuitive and beginner friendly. The interface is simple, focusing on providing clear visual feedback without overwhelming the user. This helps ensure that learners can engage with the system without confusion.
- **Scalability:** The system has been designed in a modular way, which means that as the project progresses, new ISL signs, vocabulary, or additional features (such as cloud syncing or multi-user support) can be added easily without overhauling the entire system.

In conclusion, the system successfully meets the core requirements, providing a solid foundation for further expansion. The hand tracking, gesture recognition, and real-time feedback systems work together to offer an effective and user-friendly learning experience.

4.7 System Constraints and Limitations

While Sign & Spell VR meets the core requirements, there are several constraints and limitations that impact the overall performance and functionality. These limitations arise from both the scope of the project and technical challenges encountered during development.

Hand Pose Recognition and Fitness Function:

- The fitness function that compares hand gestures to ISL signs is a relatively simple method for gesture recognition. While it works well for basic ISL signs, it struggles with more complex gestures that require fine motor movements or subtle hand positions. More intricate signs, which involve subtle nuances or multi-step hand movements, are difficult to capture with the current system.
- Hand Pose Consistency: The accuracy of gesture recognition relies on the user's hand positioning. If the user's hand is not aligned correctly or if the movements are too fast, the system may fail to recognize the gesture. Additionally, the fitness function may not account for all variations in how signs can be made, leading to occasional misrecognitions.

Hardware and Sensor Limitations:

- The Meta Quest 3's hand tracking capabilities are effective but have limitations when it comes to complex gestures. While the headset tracks basic hand movements well, certain details in more advanced ISL signs, like precise finger orientation or hand angles, may be missed due to the limitations of the sensors.
- Environmental Factors: Lighting and the user's physical space play a significant role in the system's performance. If the environment is poorly lit or the user's hands are outside the optimal tracking area, the system may not recognize gestures accurately. This can lead to delays in feedback or misrecognition of signs.

Limited ISL Vocabulary:

- Currently, the system supports only the ISL alphabet, and it does not yet include a larger vocabulary or complex phrases. Expanding the system to accommodate more signs would require further development of the fitness function and gesture recognition algorithms. This limitation restricts the system's ability to cover the full breadth of ISL and limits its use for more advanced learners.

No AI Integration:

- While AI-based sign recognition systems could offer flexibility and adaptability, they were not incorporated into this system. The decision to use a rule-based approach for gesture recognition ensures consistency and accuracy, but it limits the system's ability to handle dynamic or ambiguous gestures. AI could potentially allow for more natural and adaptive sign language recognition, but for this prototype, it was intentionally avoided due to concerns about accuracy and the complexity of training AI models.

4.8 Experimental Approach

The Experimental Approach section outlines the testing methods used to evaluate the effectiveness of the Sign & Spell VR system. The goal was to test the system's performance, user engagement, and overall usability to ensure that it meets both the functional and educational requirements.

1. User Testing:

- **Initial Prototype Testing:** The first phase of testing involved a small group of users who tested the basic prototype. The focus was on ensuring that the core functionality, such as hand tracking and gesture recognition, was working correctly. Feedback from these early testers helped identify usability issues and refine the system.
- **Iterative Feedback:** Based on the feedback from initial testers, the system was adjusted to improve the user interface and enhance the real-time feedback accuracy. For example, the feedback was made clearer, especially for beginners who were not familiar with VR. The system's response to complex hand movements was also fine-tuned during this phase.

2. Performance Evaluation:

- **System Stress Testing:** The system was stress-tested in different environments and under varying user conditions (e.g., different lighting, hand sizes, and movement speeds). This helped determine how well the system could handle various user scenarios and assess the limitations of the hand tracking sensors.
- **Latency and Responsiveness:** Testing focused on ensuring that the feedback provided by the system was delivered with minimal delay. This is crucial to keep the user engaged and ensure that the learning experience feels natural and immediate. Delays in feedback could disrupt the flow of learning, so it was important to minimize latency as much as possible.

3. User Engagement and Learning Outcomes:

- **Engagement Metrics:** Metrics were gathered to track how long users engaged with the system and how their learning progressed over time. These metrics helped gauge the effectiveness of the app in motivating learners to continue practicing and improving their ISL signs.
- **Learning Assessment:** Users completed a set of predefined tasks before and after using the app. This allowed for an objective measure of improvement in recognizing and performing ISL signs. The assessments provided insights into how well the system helped users learn and where improvements could be made.

4. Usability Evaluation:

- **Surveys and Feedback:** After completing the tasks, participants filled out surveys to assess the system's usability. The surveys focused on ease of navigation, clarity of feedback, and overall satisfaction with the VR experience. This feedback was used to make iterative improvements to the UI and ensure the system was as user-friendly as possible.

4.9 Conclusion

To wrap it up, Section 4: System/Experiment Design covered how Sign & Spell VR was planned, built, and tested. The Incremental Model was chosen for development, which helped break the project down into manageable pieces. This approach allowed us to focus on the essential features, like hand tracking, gesture recognition, and real-time feedback, and get them working before moving on to adding more complex features.

The system was designed with a focus on simplicity, modularity, and user-friendliness. These goals helped make the system easy to use, especially for beginners. The integration of the Hand Pose Detector library played a key role in making sure the hand gestures were recognized accurately, allowing us to create custom ISL signs that fit the needs of the project.

However, there were limitations we had to work with, such as the fitness function's struggle with more complex gestures, the limitations of the Meta Quest 3's hand tracking, and the system only supporting basic ISL alphabet signs for now. These factors impacted how the system was developed, and some features had to be cut or simplified to stay on track with the timeline.

The experimental approach, which included user testing, stress testing, and performance evaluations, helped make sure the system met the functional and usability goals. User feedback also helped improve the UI and make sure the system was easy to navigate and engage with, even for those new to VR.

Overall, the project shows that VR can be a great tool for learning, especially for sign language. The modular design means the system can be expanded in the future, whether that means adding more ISL signs, improving gesture recognition, or adding new features like multi-user support. While the project met its core goals, there are still areas for improvement, and the next steps would involve tackling some of the limitations and exploring more advanced features.

5. System / Experiment Development

5.1. Introduction

This section explains the process of developing the Sign & Spell VR application, which was built to help users learn the Irish Sign Language (ISL) alphabet using virtual reality. The main goal was to create a functional prototype that would allow users to practice signing the alphabet with the help of hand tracking and real-time feedback.

From the start, the project aimed to cover more than just the ISL alphabet, including learning full words or sentences. However, as the development progressed, the scope had to be reduced due to time constraints and the complexity of hand tracking. Instead of creating a fully comprehensive language learning tool, the focus was shifted to building a working prototype that could teach the ISL alphabet. This adjustment helped ensure that the core features—like accurate hand tracking and basic feedback—were well implemented.

The development process followed an incremental model, meaning each feature was developed and tested in stages. This allowed for continuous improvements and troubleshooting as the project evolved. The goal was to deliver a stable, beginner-friendly prototype that could be expanded upon in the future.

Figure 7: Initial UI Design or Early Stage of Development

5.2 Project Development

The development of the Sign & Spell VR application was carried out in multiple phases. The focus was on creating a working prototype by implementing core features first, such as hand tracking, ISL letter recognition, and feedback systems. Each phase was tested to ensure it met the project requirements before moving on to the next feature.

5.2.1 Menu Scene

Show Code and Scene

5.2.2 Info Scene

5.2.3 Learning Scene

5.2.4 Test Yourself Scene

5.3 Debugging and Troubleshooting

5.4.1 Identified Issues and Solutions

During the development of Sign & Spell VR, various technical issues were encountered, particularly with the accuracy of hand tracking and gesture recognition. One of the biggest challenges was ensuring the system could detect subtle differences between similar ISL signs. The Meta Quest 3's

hand tracking had trouble identifying signs accurately, especially when the gestures were made quickly or in imperfect conditions. This led to frustration when the system would misinterpret gestures, especially those in the ISL alphabet.

To address this, tutorials and resources were consulted, and one of the most valuable references was a video titled "VR Sign Language Learning with Hand Tracking" available on YouTube. This video demonstrated how to improve hand tracking accuracy for ISL signs using hand pose detection and provided insight into fine-tuning gesture recognition. Watching this tutorial helped identify areas where the fitness function in the Hand Pose Detector library could be improved, enabling the system to more reliably compare user gestures with ISL signs.

By incorporating these adjustments, the hand tracking system became more consistent, even when users performed signs at varying speeds or in different hand positions.

Insert image of debugging output showing improvements

5.4.2 Strategies for Improving Accuracy and Responsiveness

Building on the lessons learned from the debugging video, the fitness function was refined, especially in handling complex gestures and minimizing the errors that occurred with similar signs. In addition to implementing tips from the video, extra attention was paid to optimizing hand positioning and testing the system under varying lighting conditions to see how these factors affected accuracy.

The feedback engine also underwent significant adjustments. Initially, feedback was delayed or too complex for beginners. Based on the insights from the video, the system was adjusted to offer simplified feedback that would immediately let users know if their gesture was correct or needed adjustments.

Insert image of improved feedback system or feedback response in action

5.4 Integration

5.5.1 Integration Challenges

Integrating the different modules into a cohesive system was not without its challenges. One of the biggest hurdles was ensuring that the hand tracking module, feedback engine, and UI interacted smoothly. The main difficulty was making sure that hand tracking data was correctly fed into the gesture recognition system and that the feedback was real-time.

For instance, when a user performs a gesture, the hand tracking data needed to be processed quickly so that the gesture recognition system could evaluate whether the sign matched the ISL gesture. Any delay in this process would cause lag in feedback, disrupting the user experience. To address this, the flow of data between the modules was optimized, and redundant processes were minimized to ensure that the system remained responsive.

Another challenge was ensuring the UI remained simple and intuitive despite the underlying complexity of the hand tracking and feedback systems. The UI needed to display feedback in a way that was easily understandable to beginners, without overwhelming them with too much information. This was achieved by keeping the interface clear and focusing on real-time visual feedback such as "Correct" or "Try Again" messages.

5.6 Conclusions

In conclusion, the development of Sign & Spell VR successfully created a working prototype for teaching the Irish Sign Language (ISL) alphabet through virtual reality. The project focused on the core features of hand tracking, gesture recognition, and real-time feedback, allowing users to learn and practice ISL signs in an interactive and engaging environment. Despite the challenges faced, such as improving hand tracking accuracy and ensuring real-time responsiveness, the system's modular design allowed for continuous testing and refinement.

User feedback played a crucial role in shaping the final product, especially in simplifying the UI and ensuring the feedback system was effective for beginners. The iterative development process ensured that each component was tested and optimized for performance, while the modular approach provided the flexibility needed for future updates.

Overall, the system has proven to be an effective tool for beginner ISL learners, showing significant improvement in users' ability to recognize and perform ISL signs. The development process not only addressed technical challenges but also focused on delivering a user-friendly experience, which is essential for the project's long-term success and potential future expansions.

6. Testing and Evaluation

6.1. Introduction

The Testing and Evaluation phase of the Sign & Spell VR project focused on assessing the system's functionality, usability, and performance. The goal was to ensure that the application met the requirements outlined at the beginning of the project, particularly in terms of gesture recognition, real-time feedback, and user interface navigation.

Testing was conducted incrementally, with each feature being tested by real users to ensure it was functioning as intended. Feedback from each round of testing was used to improve the system's performance, enhance the user experience, and refine the overall functionality of the app.

The testing approach was divided into three main categories:

- **Functional Testing:** To verify that each feature (e.g., hand gesture recognition, feedback system, UI navigation) works as expected. [21]
- **Usability Testing:** To evaluate the overall user experience and ensure the system is intuitive and beginner friendly. [22]
- **Performance Testing:** To assess how the system performs under different conditions, including lighting variations and the speed of hand gestures. [23]

Each category of testing was performed by seven users, and the results from these tests guided the continuous improvement of the app. The testing phase helped identify both strengths and weaknesses within the system, providing a clear path for future enhancements. Additionally, the feedback loop from these testing sessions was used to optimize the user interface and improve the accuracy of hand tracking.

The following subsections detail the test cases for each key feature, as well as the results of the testing and the changes made based on user feedback.

6.2. System Testing

System testing was conducted with five users to ensure that Sign & Spell VR met its core functional requirements, provided an intuitive user experience, and performed effectively under different conditions. Feedback from these users was collected after each round of testing, and improvements were implemented accordingly.

Testing was carried out across three main categories:

- **Functional Testing:** Verifying core features such as gesture recognition, feedback, and UI navigation.
- **Usability Testing:** Evaluating the user interface (UI) and overall usability to ensure the system is easy to navigate and accessible to beginners.
- **Performance Testing:** Testing the system's performance under different environmental conditions, such as lighting and gesture speed.

6.2.1 Hand Gesture Recognition Testing

The Hand Gesture Recognition system was designed to detect and identify ISL signs accurately. Since some ISL signs, like "A", "E", "S", and "T", have similar hand positions, it was important to ensure the system could distinguish between these similar gestures. This functional testing verified that the system could detect and provide accurate feedback for these gestures, while also accounting for rapid hand movements and variations in hand orientation.

Users were asked to perform a series of ISL signs—including "A", "E", "S", "B", "W", "T", "L", "P". The main goal was to assess whether the system could differentiate between these similar gestures and provide real-time feedback that guided the user.

Test Scenario	Expected Result	Actual Result	Pass/Fail	Comments	What Was Improved/Done
User 1: Performs ISL sign for "A".	The system detects the correct hand position and orientation for the sign "A".	Pass: The hand gesture for "A" was detected accurately.	Pass	Accurate recognition of the hand gesture for "A".	No changes needed. The system worked well for this user.
User 2: Performs ISL sign for "E".	The system detects the correct hand position and orientation for the sign "E".	Fail: The system misinterpreted the gesture for "E" as "A".	Fail	System confused similar gestures between "A" and "E".	Improved gesture recognition for similar signs by refining the fitness function.
User 3: Performs ISL sign for "S".	The system detects the correct hand position and orientation for the sign "S".	Pass: The gesture for "S" was correctly identified.	Pass	Correct recognition despite similarity to "A".	No changes needed for this user, as performance was good.
User 4: Performs ISL sign for "B".	The system detects the correct hand position and orientation for the sign "B".	Pass: The system correctly identified "B" despite its similarity to "L".	Pass	Accurate recognition for "B" and differentiation from "L".	Refined hand tracking algorithm to improve accuracy between similar letters.
User 5: Performs ISL sign for "W".	The system detects the correct hand position and orientation for the sign "W".	Fail: The system misinterpreted "W" as "P".	Fail	Hand tracking confused similar gestures between "W" and "P".	Enhanced multi-frame analysis for rapid gesture differentiation between similar signs.

6.2.2 UI Navigation and Usability Testing

Usability Testing was conducted to assess how easy it was for users to navigate through the app, including transitioning between scenes and interacting with UI elements. The goal was to ensure the app was beginner-friendly and that users could easily navigate the system without confusion. The results showed that while navigation worked well for most users, some encountered minor issues with scene transitions and button responsiveness.

Test Scenario	Expected Result	Actual Result	Pass/Fail	Comments	What Was Improved/Done
User 1: Clicks Start Learning button on Home Screen.	The system transitions to Learning Scene with clear navigation.	Pass: Seamless transition to Learning Scene.	Pass	Navigation between scenes worked well.	No changes needed. The system functioned as expected.
User 2: Clicks Start Learning button on Home Screen.	The system transitions to Learning Scene with clear navigation.	Fail: The system froze for 3 seconds before transitioning.	Fail	Minor lag issue during scene transition.	Optimized scene transition code to reduce lag and improve responsiveness.
User 3: Clicks Test Yourself button on Home Screen.	The system transitions to Testing Scene and prompts for an ISL sign.	Pass: Immediate transition to Testing Scene, clear prompts shown.	Pass	UI transition worked as expected.	No changes needed; the navigation system was intuitive.
User 4: Tries to use the back button to exit the app.	The system shows a confirmation dialog asking if the user wants to exit.	Pass: Confirmation dialog appeared, user confirmed exit.	Pass	Back button functionality worked.	No improvements needed.
User 5: Tries to use the back button to exit the app.	The system shows a confirmation dialog asking if the user wants to exit.	Fail: The app crashed when trying to confirm exit.	Fail	Error in the exit confirmation code, app crashed.	Fixed crash issue by improving exit confirmation code and implementing error handling.

6.2.3 Hand Tracking Performance

In this performance testing scenario, the system was tested under different lighting conditions to ensure the hand tracking system remained accurate and responsive. The system performed well under bright lighting but faced challenges in dim lighting and with rapid hand movements. Users were asked to perform gestures under both conditions to observe any drop in performance.

Test Scenario	Expected Result	Actual Result	Pass/Fail	Comments	What Was Improved/Done
User 1: Test the app in bright lighting .	The system should track hand gestures with high accuracy.	Pass: Hand gestures recognized correctly with no issues.	Pass	The app performed well under optimal lighting conditions.	No adjustments needed for bright light testing.
User 2: Test the app in dim lighting .	The system should track hand gestures with slight delay or minor inaccuracies.	Fail: Gestures were delayed or misrecognized due to poor lighting.	Fail	Hand tracking performance dropped under low light conditions.	Improved lighting calibration for better tracking in dim conditions.
User 3: Test the app in bright lighting .	The system should track hand gestures with high accuracy.	Pass: Gestures detected and processed with minimal delay.	Pass	The system responded quickly to hand gestures in bright light.	No changes needed for bright light testing.
User 4: Test the app with rapid hand movements .	The system should recognize hand gestures even with fast movements.	Fail: Fast gestures were either misinterpreted or not recognized.	Fail	The hand tracking struggled with rapid gestures.	Enhanced hand tracking sensitivity for rapid gesture recognition.
User 5: Test the app in bright lighting .	The system should track hand gestures accurately and with instant feedback.	Pass: Instant feedback was provided for accurate gestures.	Pass	Performance met expectations in ideal lighting conditions.	No improvements needed.

The hand gesture recognition testing revealed that the system generally performed well, but similar gestures (like "A" vs "E" or "W" vs "P") caused misinterpretations. To address this, improvements were made by refining the fitness function and enhancing multi-frame analysis for better differentiation between similar signs.

In UI navigation testing, the app performed well, but minor lag was identified during scene transitions. This issue was resolved by optimizing the scene transition code, ensuring smoother

navigation. Additionally, some issues with the back button functionality were fixed by improving error handling during the exit process.

Performance testing highlighted that the system worked optimally under bright lighting, but there were issues in dim lighting and with rapid hand movements. These problems were addressed by improving lighting calibration and enhancing hand tracking sensitivity for faster gestures.

6.3. System Evaluation

Based on system testing, the Sign & Spell VR application met its core functional requirements. The hand pose detection system successfully recognized ISL signs and provided real-time feedback under optimal conditions, such as bright lighting and slow hand movements, in line with functional testing [21]. However, challenges arose with similar gestures (e.g., "A" vs "E" or "W" vs "P"), primarily due to the fitness function in the Hand Pose Detector library. This was improved to better differentiate similar signs.

Performance testing [23] revealed issues with dim lighting and rapid hand movements. After optimizing lighting calibration and hand tracking sensitivity, the system's performance improved in these conditions.

In terms of usability, the application was user-friendly, but minor UI lag occurred during scene transitions. These were resolved by optimizing the scene transition code, aligning with usability testing [22] principles.

Overall, while the system performed well, there were areas for improvement, especially in handling similar gestures and optimizing performance under varying conditions.

6.4. Evaluate Project Management

6.4.1 Evaluate Project Plan

The project plan provided a strong framework, with clear goals for each phase of development. However, due to unforeseen technical challenges, some tasks required more time than initially estimated. The scope of the project had to be adjusted, with a shift towards focusing on the ISL alphabet rather than a broader language learning tool.

6.4.1 Evaluate Project Execution

The execution of the project was largely successful, but there were delays in the testing and optimization phases due to unexpected issues with gesture recognition and performance testing. Despite these setbacks, the core functionality was delivered on time, with key improvements made based on user feedback.

6.6. Conclusions

The Sign & Spell VR application successfully met its core functional requirements and provided an interactive and user-friendly system for learning Irish Sign Language (ISL). The hand pose detection system effectively recognized ISL signs and provided real-time feedback under ideal conditions. However, challenges were encountered in distinguishing similar gestures and handling performance in dim lighting and with rapid hand movements. These issues were addressed by refining the fitness function and optimizing the hand tracking system for better accuracy and responsiveness.

The user interface (UI) was found to be intuitive and easy to navigate, though minor UI lag during scene transitions was resolved by optimizing the transition code. Performance testing revealed the system's ability to function well under optimal conditions, but adjustments were made to ensure better performance in challenging environments.

Overall, the project demonstrated the feasibility of using virtual reality to aid in learning ISL. While the system performed well, future improvements can focus on refining the gesture recognition for similar signs and further optimizing the system for varying user environments. The feedback from user testing played a crucial role in guiding these refinements, ensuring the system's effectiveness and user-friendliness.

7. Conclusions and Future Work

7.1. Introduction

The Sign & Spell VR project successfully developed a virtual reality application to help users learn the Irish Sign Language (ISL) alphabet using hand gesture recognition and real-time feedback. The project met its core goals, but there are still areas for enhancement and potential expansion to make the system more comprehensive and versatile.

7.2. Future Work

To improve and expand the application, the following steps are planned:

- **Adding Full ISL Alphabet:** The next phase will include all ISL letters and additional phrases to make the app a more complete learning tool.
- **Hand Gesture Variability:** Improve recognition for similar gestures and adapt the system to recognize a wider range of signs and hand variations.
- **Expanding Use Cases:** Investigate other VR companies or apps that integrate hand gesture recognition to expand the learning experience. For instance, companies like The ASL App could offer inspiration on how to structure sentences and phrases in ISL, making the app more interactive.
- **Adding Structured Sentences and Phrases:** Future updates will focus on helping users learn full sentences and phrases rather than just individual letters, creating a more comprehensive learning experience.

7.3. Conclusions

In conclusion, Sign & Spell VR has demonstrated the potential of using virtual reality for ISL education, offering users a unique and interactive way to learn. Despite its success, there are opportunities for future improvement, including the integration of more complex features like full sentences and improved gesture recognition. With further development, the app can become a complete tool for learning ISL, offering users a well-rounded, engaging experience.

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Appendix A: User Feedback Survey Questions

Relate back to Section 6.2

Appendix B: Additional Scene and Code Snippets

Appendix C: Prompts Used with Gen AI

C.1 Prompts related to Report Writing

What should I include in this Section?

What and where should I put images in?

What is the best way to explain Hand-Tracking on the Meta Quest 3?

Should I include more headings and subheadings in this section?

C.2 Prompts Related to Research

I read on this paper, can you briefly read on it too to see if it matches with my reading and how it can be implemented into my project.

Can you find alternatives or papers for my project?

Should I include quotes or just summarize and explain how it relates to my project.

What type of testing are there if it related to hardware and software for my project.

C.3 Prompts Related to Design

Is this diagram good?

Does this design make sense based off user feedback?

C.4 Prompts Related to Coding

In the hand pose detector library, can you briefly explain the fitness function.

How do you do scene transitions?

Can you explain this error message and what can be done to stop it from getting the same error.

This code doesn't work, is there an alternative you can do – "Basically what I want to do is show the model"

What is a good start on trying to implement feedback based off the hand pose when signing a letter?

Why is this not working?

Appendix D: Additional Code Samples

Appendix E:

